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Riemann–Hilbert Approach for the Hirota–Satsuma Coupled KdV System

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ABSTRACT

In this paper, the Riemann–Hilbert (RH) method is developed to solve the Hirota–Satsuma coupled KdV (HScKdV) system associated with 4×4 matrix spectral problem. Because the spectral matrix is asymmetric and high order, it brings great difficulties to analysis and solution. First, a direct scattering problem is carried out, from which the initial data are mapped to the scattering data. On the basis of introducing the generalized cross product of vectors, the basic meromorphic matrix eigenfunctions of corresponding Lax pairs are expressed by the Jost solutions and adjoint Jost solutions. Then, the inverse scattering problem is characterized as the 4×4 matrix RH problem, which gives the formula for constructing solutions of the HScKdV system. As an example, the RH problem corresponding to the reflectionless case is solved and the soliton solution of the HScKdV system is obtained.

MATHEMATICS SUBJECT CLASSIFICATION: 35Q15, 35B40

1 | Introduction

The Hirota–Satsuma coupled KdV (HScKdV) system [1, 2],

$$\begin{cases} u_t = \frac{1}{2}u_{xxx} - 3uu_x + 6vv_x, \\ v_t = -v_{xxx} + 3uv_x, \end{cases} \tag{1.1}$$

can be used to describe the interaction of two long waves with different dispersion relations, which is reduced to the KdV equation [3, 4] when $v = 0$. System (1.1) has been studied extensively since it was proposed. Soliton solutions of the HScKdV system are obtained by using the bilinear method [1]. It is also showed that the HScKdV system is a four-reduction of the well-known Kadomtsev–Petviashvili hierarchy [2]. Dodd showed this system is completely integrable in the sense of

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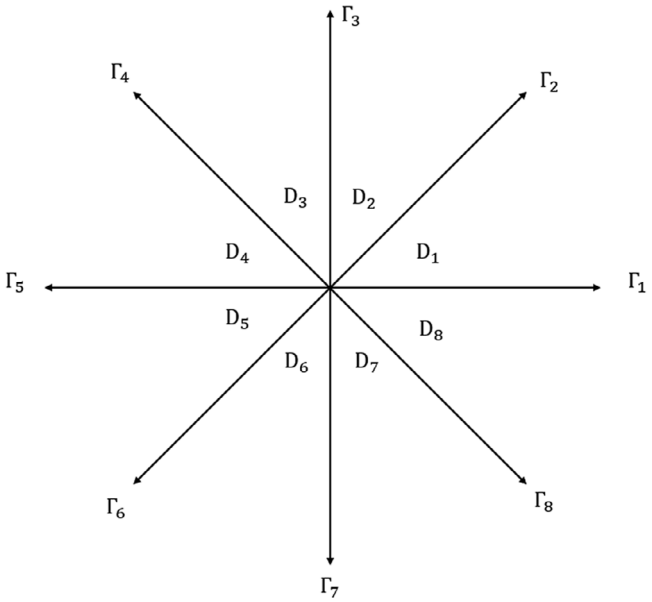


FIGURE 1 | The contour $\Gamma = \cup_{j=1}^{8}\Gamma_j$ and the open sets $\{D_j\}_{j=1}^{8}$.

having a Lax pair [5]. More exact solutions under nonzero background are obtained by Tam et al. [6]. Levi constructed the Bäcklund transformation for this system [7]. Hu and Liu gave its binary Darboux transformation [8]. A three-component generalization of this system is given in Ref. [9]. To the best of our knowledge, no one has performed inverse scattering analysis on this system so far. The aim of this work is to fill this gap.

Solving integrable nonlinear partial differential equations by the inverse scattering transform (IST) is an important milestone in soliton theory. The modern version of the IST method usually relies on dealing with the associated Riemann–Hilbert (RH) problem [10], so it is also called the RH method. The advantage of this method is that once the relevant RH problem is established, not only the soliton solution can be obtained in reflectionless case, but also the Deift–Zhou steepest descent method [11] can be applied to study the asymptotic behavior of the solutions [12–14]. This is one of our main motivations for deriving RH representations for solutions of the HScKdV system. We hope to analyze the long-time asymptotic behavior of the solution for the HScKdV system in the following work.

The main difficulties we encountered in this paper are as follows. It is well-known that in the inverse scattering analysis, in order to correctly recover the solutions of the associated soliton equation, it is necessary to fully characterize the symmetries of the eigenfunctions and the scattering data. Therefore, the corresponding Lax pair is required to have a "good" symmetry. We first observe that the Lax pair of system (1.1) (see (2.1)) is very irregular and not suitable for inverse scattering analysis. Therefore, we need to find a gauge transformation (see (2.3)) that converts the Lax pair to the Zakharov–Shabat-type spectral problems. Since the transformed Lax pair has a more complex form, it is very difficult to find a suitable symmetry. But fortunately, after a detailed calculation, we finally found three suitable symmetries, which allowed our inverse scattering analysis to proceed smoothly. Note that the Lax pair of the HScKdV system is fourth order, which makes the inverse scattering analysis extremely difficult. In the low-order case (such as the nonlinear Schrödinger [NLS] equation), we need to define the fundamental matrix meromorphic eigenfunctions of the corresponding Lax pair in the upper half complex plane and the lower half complex plane, and use them to derive the related RH problem. Such fundamental matrix eigenfunction solution is called the Beals–Coifman simultaneous solution of the corresponding Lax pair [15]. One usually use the Jost solutions of the Lax pair to define the Beals–Coifman simultaneous solution, because there are two columns of analytic Jost solutions in each half-plane. Different from the low-order case, the construction of the fundamental meromorphic eigenfunction solution of the high-order Lax pair becomes more complicated. As one will see, in our case, we need to define a fundamental matrix eigenfunction in each of the eight regions $\{D_j\}_{j=1}^{8}$ (see Figure 1). However, we cannot use only the Jost solutions to achieve our goal as in the low-order case, because there are only two columns of analytical Jost solutions in each region D_j. In order to overcome this difficulty, we have to construct two additional auxiliary solutions in each region. Such constructions have been shown in Refs. [16–18], where tensor method is applied to implement this process. However, in this paper, we will follow another idea to complete the construction of the fundamental matrix eigenfunctions, which have been used in Refs. [19–29] to construct fundamental matrix eigenfunctions of the associated 3×3 Lax pairs. This method is to use a set of Fredholm integral equations to define the fundamental matrix eigenfunctions directly in each region, and then establish the relationship between them and the

Jost solutions. Compared with the 3×3 matrix Lax system, it becomes more complicated to establish the relationship between the Jost solutions and the fundamental matrix eigenfunctions in the 4×4 case. In order to achieve this goal, we have to introduce the fundamental tensors and a generalized version of the familiar cross product [16, 18]. Our results show that the auxiliary solutions we need are actually the generalized cross products of some adjoint Jost solutions. These representations are similar to those obtained by the tensor method in the context of the three-component defocusing NLS equation [18].

For convenience, we restrict our attention to the potentials that decay rapidly as $|x|\to\infty$. In other words, our inverse scattering analysis is based on Schwarz class solutions of system (1.1) (note that the soliton solutions are included). The so-called Schwartz class solutions of (1.1) are defined as follows:

Definition 1.1. We call $\{u(x,t),v(x,t)\}$ a Schwartz class solution of (1.1) with existence time $T\in(0,+\infty]$ and initial data $u_0,v_0\in\mathcal{S}(\mathbb{R})$ if (i) u and v are smooth real-valued functions of $(x,t)\in\mathbb{R}\times[0,T)$; (ii) u and v satisfy (1.1) for $(x,t)\in\mathbb{R}\times[0,T)$ and

$$u(x,0)=u_0,\qquad v(x,0)=v_0,\qquad x\in\mathbb{R};$$

(iii) u and v have rapid decay as $|x|\to\infty$ in the sense that, for each integer $N\geq 1$

$$\sup_{x\in\mathbb{R}}\sup_{t\in[0,T)}\sum_{j=0}^{N}(1+|x|)^N\left(\left|\partial_x^j u\right|+\left|\partial_x^j v\right|\right)<+\infty.$$

We establish the inverse scattering analysis in two special cases: (1) soliton-free case; and (2) pure soliton case. In the first case, it is assumed that there is no discrete spectrum, and the scattering matrix $s(k)$ is required to have a general asymptotic behavior as $k\to 0$ (see Assumptions 3.1 and 3.2). Under these assumptions, we find that the reflection coefficients have nice properties, and we summarize them as Theorem 3.1. Then, we define a suitable matrix RH problem based on these reflection coefficients. In Theorem 3.2, we establish the relationship between the solution of the RH problem and the potentials $\{u,v\}$. In the second case, Assumptions 3.1 and 3.2 are broken, so we have to define another matrix RH problem to derive the soliton solutions. By solving our RH problem, we can obtain the exact expression of the soliton solutions. As an example, we show the 1-soliton solutions. And we find that they have the same form as those obtained in Ref. [1].

The outline of this work is as follows. In Section 2, starting from the initial data $\{u_0,v_0\}$, we define the Jost solutions, the scattering matrix, and the fundamental meromorphic eigenfunction. In Section 3, we construct RH problems in two cases, and reconstruct the potentials $\{u,v\}$ in terms of the solutions of these RH problems.

2 | Spectral Analysis

2.1 | Lax Pair

The HScKdV system (1.1) admits the following Lax pair [9]:

$$\psi_x=U\psi,\qquad \psi_t=V\psi,\tag{2.1}$$

where $\psi(x,t,k)$ is a 4×4 matrix-valued function, $k\in\mathbb{C}$ is the spectral parameter, and the matrices $U(x,t,k)$ and $V(x,t,k)$ are given by

$$U=U^{\infty}+U_1,\quad V=V^{\infty}+V_1,\tag{2.2}$$

with

$$U^{\infty}=\begin{pmatrix}0&0&1&0\\0&0&0&1\\k^2&0&0&0\\0&-k^2&0&0\end{pmatrix},\quad U_1=\begin{pmatrix}0&0&0&0\\0&0&0&0\\u&v&0&0\\v&u&0&0\end{pmatrix},$$

$$V^{\infty}=\begin{pmatrix}0&0&2k^2&0\\0&0&0&-2k^2\\2k^4&0&0&0\\0&2k^4&0&0\end{pmatrix},\quad V_1=\begin{pmatrix}\frac{u_x}{2}&-v_x&-u&2v\\-v_x&\frac{u_x}{2}&2v&-u\\g_1&-v_{xx}+uv&-\frac{u_x}{2}&v_x\\-v_{xx}+uv&g_2&v_x&-\frac{u_x}{2}\end{pmatrix},$$

$$g_1=\frac{u_{xx}}{2}+2uv-u^2+k^2u,\qquad g_2=\frac{u_{xx}}{2}+2uv-u^2-k^2u.$$

It is easy to verify that (1.1) is the compatibility condition of (2.1) by direct calculation. Since $\{u,v\}$ is a Schwartz class solution of (1.1), U_1 and V_1 decay rapidly as $|x|\to\infty$.

Now we introduce a gauge transformation

$$\psi(x,t,k)=P(k)\phi(x,t,k),\quad P(k)=\begin{pmatrix}0&1&0&1\\1&0&1&0\\0&-k&0&k\\\mathrm{i}k&0&-\mathrm{i}k&0\end{pmatrix}\tag{2.3}$$

to diagonalize the matrices U^{∞} and V^{∞}, where $\mathrm{i}^2=-1$. Then, the Lax pair (2.1) is transformed into

$$\phi_x=F\phi,\qquad \phi_t=G\phi,\tag{2.4}$$

where

$$F(x,t,k)=P^{-1}(k)U(x,t,k)P(k)=kJ+P^{-1}(k)U_1(x,t)P(k),\tag{2.5}$$

$$G(x,t,k)=P^{-1}(k)V(x,t,k)P(k)=2k^3\Omega+P^{-1}(k)V_1(x,t,k)P(k),\tag{2.6}$$

with

$$J=\mathrm{diag}(\mathrm{i},-1,-\mathrm{i},1),\qquad \Omega=J^3=\mathrm{diag}(-\mathrm{i},-1,\mathrm{i},1).$$

Moreover, by a direct calculation we find

$$U_2=P^{-1}U_1P=\frac{u}{2k}\begin{pmatrix}-\mathrm{i}&0&-\mathrm{i}&0\\0&-1&0&-1\\\mathrm{i}&0&\mathrm{i}&0\\0&1&0&1\end{pmatrix}+\frac{v}{2k}\begin{pmatrix}0&-\mathrm{i}&0&-\mathrm{i}\\-1&0&-1&0\\0&\mathrm{i}&0&\mathrm{i}\\1&0&1&0\end{pmatrix},\tag{2.7}$$

$$V_2=P^{-1}V_1P\to 0\quad\text{rapidly},\qquad \text{as}|x|\to\infty.\tag{2.8}$$

Note that the determinant of $P(k)$ is $-4\mathrm{i}k^2$, so the above transformation holds only when $k\in\mathbb{C}\setminus\{0\}$. It is not convenient to analyze the eigenfunction $\phi(x,t,k)$ directly. To this end, we consider the modified eigenfunction $\mu(x,t,k)=\phi(x,t,k)\mathrm{e}^{-L(k)x-Z(k)t}$, where $L(k)$ and $Z(k)$ are given by

$$\begin{aligned}L(k)&=kJ:=\mathrm{diag}(l_1,l_2,l_3,l_4),\\Z(k)&=2k^3\Omega:=\mathrm{diag}(z_1,z_2,z_3,z_4).\end{aligned}\tag{2.9}$$

Then according to (2.4), $\mu(x,t,k)$ satisfies

$$\begin{cases}\mu_x-[L,\mu]=U_2\mu,\\\mu_t-[Z,\mu]=V_2\mu.\end{cases}\tag{2.10}$$

Next let us discuss the symmetries of $F(x,t,k)$ and $G(x,t,k)$. The following results can be obtained by direct calculation.

Lemma 2.1. *The matrix-valued functions $F(x,t,k)$ and $G(x,t,k)$ defined by (2.5) and (2.6) satisfy*

$$AF(\mathrm{i}k)A^{-1} = F(k), \qquad AG(\mathrm{i}k)A^{-1} = G(k), \tag{2.11a}$$

$$BF^*(k^*)B = F(k), \qquad BG^*(k^*)B = G(k), \tag{2.11b}$$

where

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{2.12}$$

2.2 | Jost Solutions $\mu_\pm(x,k)$

Since the time dependence of the potentials do not play any role in the direct problem, we fix $t = 0$ until Section 3. We abbreviate $U_2(x,0,k)$ as $U_2(x,k)$. Now the x-part of (2.10) is rewritten as

$$\mu_x(x,k) - [L(k), \mu(x,k)] = U_2(x,k)\mu(x,k). \tag{2.13}$$

The Jost solutions $\mu_\pm(x,k)$ are defined as the solutions of the above equation satisfying boundary conditions

$$\mu_\pm(x,k) = I + o(1), \qquad x \to \pm\infty, \tag{2.14}$$

or equivalently they can be defined by the following integral equations:

$$\mu_\pm(x,k) = I + \int_{\pm\infty}^{x} \mathrm{e}^{(x-y)\hat{L}(k)}(U_2\mu_\pm)(y,k)dy, \tag{2.15}$$

where $\mathrm{e}^{(x-y)\hat{L}(k)}(\cdot) = \mathrm{e}^{(x-y)L(k)}(\cdot)\mathrm{e}^{-(x-y)L(k)}$. In order to describe the properties of Jost solutions $\mu_\pm(x,k)$, we need to introduce some notations. We decompose the complex k-plane into the eight open subsets $\{D_n\}_1^8$ (see Figure 1) defined by

$$D_1 = \{k \in \mathbb{C} \mid \mathrm{Re}l_4 > \mathrm{Re}l_3 > \mathrm{Re}l_1 > \mathrm{Re}l_2\}, \quad D_2 = \{k \in \mathbb{C} \mid \mathrm{Re}l_3 > \mathrm{Re}l_4 > \mathrm{Re}l_2 > \mathrm{Re}l_1\},$$

$$D_3 = \{k \in \mathbb{C} \mid \mathrm{Re}l_3 > \mathrm{Re}l_2 > \mathrm{Re}l_4 > \mathrm{Re}l_1\}, \quad D_4 = \{k \in \mathbb{C} \mid \mathrm{Re}l_2 > \mathrm{Re}l_3 > \mathrm{Re}l_1 > \mathrm{Re}l_4\},$$

$$D_5 = \{k \in \mathbb{C} \mid \mathrm{Re}l_2 > \mathrm{Re}l_1 > \mathrm{Re}l_3 > \mathrm{Re}l_4\}, \quad D_6 = \{k \in \mathbb{C} \mid \mathrm{Re}l_1 > \mathrm{Re}l_2 > \mathrm{Re}l_4 > \mathrm{Re}l_3\},$$

$$D_7 = \{k \in \mathbb{C} \mid \mathrm{Re}l_1 > \mathrm{Re}l_4 > \mathrm{Re}l_2 > \mathrm{Re}l_3\}, \quad D_8 = \{k \in \mathbb{C} \mid \mathrm{Re}l_4 > \mathrm{Re}l_1 > \mathrm{Re}l_3 > \mathrm{Re}l_2\}.$$

Let $\Gamma = \cup_{j=1}^8 \Gamma_j$ represent the boundary of regions $\{D_n\}_1^8$, which is composed of eight rays intersecting at the origin, and the direction of Γ is shown in the Figure 1. And S denotes the interior of $\bar{D}_4 \cup \bar{D}_5$, $\mathrm{i}S$ denotes the region obtained by rotating S 90° degrees counterclockwise. Similarly, one can define $-S$ and $-\mathrm{i}S$.

By analyzing the Neumann series of integral equations (2.15), we obtain the basic properties of the Jost solutions $\mu_\pm(x,k)$. Since these techniques are rather standard by now, we omit the proofs and give the following proposition directly.

Proposition 2.1. *Suppose $u_0, v_0 \in \mathcal{S}(\mathbb{R})$, then $\mu_\pm(x,k)$ defined by (2.15) have the following properties:*

a. *For each $x \in \mathbb{R}$, the function $\mu_+(x,k)$ is well-defined and continuous for $k \in (-\mathrm{i}\bar{S}, -\bar{S}, \mathrm{i}\bar{S}, \bar{S}) \setminus \{0\}$. In addition, $\mu_+(x,k)$ is analytic for $k \in (-\mathrm{i}S, -S, \mathrm{i}S, S) \setminus \{0\}$.*
b. *For each $x \in \mathbb{R}$, the function $\mu_-(x,k)$ is well-defined and continuous for $k \in (\mathrm{i}\bar{S}, \bar{S}, -\mathrm{i}\bar{S}, -\bar{S}) \setminus \{0\}$. In addition, $\mu_-(x,k)$ is analytic for $k \in (\mathrm{i}S, S, -\mathrm{i}S, -S) \setminus \{0\}$.*
c. *For each $x \in \mathbb{R}$ and positive integer j, the partial derivative $\frac{\partial^j \mu_+(x,k)}{\partial k^j}$ of $\mu_+(x,k)$ is well-defined and analytic for $k \in (-\mathrm{i}S, -S, \mathrm{i}S, S) \setminus \{0\}$, and it can be continuously extended to the boundary of its analytic region except the origin.*

d. *For each* $x \in \mathbb{R}$ *and positive integer* j, *the partial derivative* $\frac{\partial^j \mu_-(x,k)}{\partial k^j}$ *of* $\mu_-(x,k)$ *is well-defined and analytic for* $k \in (\mathrm{i}S, S, -\mathrm{i}S, -S) \setminus \{0\}$, *and it can be continuously extended to the boundary of its analytic region except the origin.*

e. $\mu_\pm(x,t,k)$ *satisfies these two symmetries:*

$$\mu_\pm(x,k) = A\mu_\pm(x,\mathrm{i}k)A^{-1}, \quad \mu_\pm(x,k) = B\mu_\pm^*(x,k^*)B, \tag{2.16}$$

where A, B *are given by (2.12).*

f. *When* u_0 *and* v_0 *has compact support, then* $\mu_\pm(x,k)$ *are analytic for* $k \in \mathbb{C} \setminus \{0\}$ *and* $\det\mu_\pm(x,k) = 1$.

Now let us characterize the asymptotic behavior of the Jost solutions when k tends to infinity. Consider the formal power series

$$\mu_\pm(x,k) = I + \frac{\mu_\pm^{[1]}}{k} + \frac{\mu_\pm^{[2]}}{k^2} + \cdots, \tag{2.17}$$

with the following normalization condition:

$$\lim_{x\to\pm\infty} \mu_\pm^{[n]} = 0, \quad n \geq 1. \tag{2.18}$$

Denote the diagonal part of a matrix A as $[A]_d$ and the nondiagonal part as $[A]_o$. Substituting the above series into (2.13) and comparing the coefficients of the same powers, we arrive at the recurrence relation:

$$\begin{cases} \left[J, \mu_\pm^{[j+1]}\right] = \left[\partial_x \mu_\pm^{[j]}\right]_o - \left[\tilde{U}_2 \mu_\pm^{[j-1]}\right]_o, \\ \partial_x \left[\mu_\pm^{[j+1]}\right]_d = \left[\tilde{U}_2 \mu_\pm^{[j]}\right]_d, \end{cases} \qquad j \geq 0, \tag{2.19}$$

where $\tilde{U}_2 = kU_2$. In addition, we need to make the initial assignments $\mu_\pm^{[-1]} = 0_{4\times4}$ and $\mu_\pm^{[0]} = I$. Combining the above with condition (2.18), we find that the formal power series (2.17) can be fully determined by the recurrence relation (2.19). The first few coefficients are given by

$$\begin{aligned} \left[\mu_\pm^{[1]}\right]_d &= \frac{\int_{\pm\infty}^x u_0(y)dy}{2} \begin{pmatrix} -\mathrm{i} & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & \mathrm{i} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \left[\mu_\pm^{[1]}\right]_o = 0_{4\times4}, \\ \left[\mu_\pm^{[2]}\right]_d &= \frac{\int_{\pm\infty}^x u_0(y) \int_{\pm\infty}^y u_0(\xi)d\xi dy}{4} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \\ \left[\mu_\pm^{[2]}\right]_o &= \begin{pmatrix} 0 & \frac{1+\mathrm{i}}{4}v_0 & \frac{1}{4}u_0 & \frac{1-\mathrm{i}}{4}v_0 \\ \frac{\mathrm{i}-1}{4}v_0 & 0 & -\frac{1+\mathrm{i}}{4}v_0 & -\frac{1}{4}u_0 \\ \frac{1}{4}u_0 & \frac{1-\mathrm{i}}{4}v_0 & 0 & \frac{1+\mathrm{i}}{4}v_0 \\ -\frac{1+\mathrm{i}}{4}v_0 & -\frac{1}{4}u_0 & \frac{\mathrm{i}-1}{4}v_0 & 0 \end{pmatrix}. \end{aligned} \tag{2.20}$$

The following proposition shows that (2.17) are good approximations of $\mu_\pm(x,k)$ as $k \to \infty$. The proof is similar to the proof of [30, Theorem 3.3] and is omitted.

Proposition 2.2. *Suppose $u_0, v_0 \in \mathcal{S}(\mathbb{R})$, then the formal power series (2.17) are exactly the asymptotic expansion of $\mu_\pm(x,t)$ as $k \to \infty$. More precisely, let p be a positive integer, then the estimates*

$$\left| \frac{\partial^j}{\partial k^j} \left(\mu_+ - I - \sum_{l=1}^{p} \frac{\mu_+^{[l]}}{k^l} \right) \right| \leq \frac{f_+(x)}{|k|^{p+1}}, \quad x \in \mathbb{R}, \quad k \in (-i\bar{S}, -\bar{S}, i\bar{S}, \bar{S}), \quad |k| \geq 2, \tag{2.21a}$$

and

$$\left| \frac{\partial^j}{\partial k^j} \left(\mu_- - I - \sum_{l=1}^{p} \frac{\mu_-^{[l]}}{k^l} \right) \right| \leq \frac{f_-(x)}{|k|^{p+1}}, \quad x \in \mathbb{R}, \quad k \in (i\bar{S}, \bar{S}, -i\bar{S}, -\bar{S}), \quad |k| \geq 2 \tag{2.21b}$$

hold for each integer $j \geq 0$, where $f_+(x)$ and $f_-(x)$ are bounded smooth positive functions of $x \in \mathbb{R}$ with rapid decay as $x \to \pm\infty$, respectively.

Next, we study the properties of the Jost solutions when k tends to 0. The fact that $U_2(x,k)$ has a simple pole at the origin suggests that the origin may be a singularity of the Jost solutions $\mu_\pm(x,k)$. The following proposition accurately characterizes asymptotic behavior of $\mu_\pm(x,k)$ in a sufficiently small neighborhoods of the origin.

Proposition 2.3. *Suppose $u_0, v_0 \in \mathcal{S}(\mathbb{R})$, and let $p \geq 0$ be an integer, then there exist fourth-order matrix-valued functions $C_\pm^{(l)}(x), \quad l = -1, 0, 1, \ldots, p$, with the following properties: The estimates*

$$\left| \mu_+ - I - \sum_{l=-1}^{p} C_+^{(l)} k^l \right| \leq g_+(x)|k|^{p+1-j}, \quad x \in \mathbb{R}, \quad k \in (-\mathrm{i}\bar{S}, -\bar{S}, \mathrm{i}\bar{S}, \bar{S}), \quad |k| \leq \frac{1}{2} \tag{2.22a}$$

and

$$\left| \mu_- - I - \sum_{l=-1}^{p} C_-^{(l)} k^l \right| \leq g_-(x)|k|^{p+1-j}, \quad x \in \mathbb{R}, \quad k \in (\mathrm{i}\bar{S}, \bar{S}, -\mathrm{i}\bar{S}, -\bar{S}), \quad |k| \leq \frac{1}{2} \tag{2.22b}$$

hold, where $g_+(x)$ and $g_-(x)$ are bounded smooth positive functions of $x \in \mathbb{R}$ with rapid decay as $x \to \pm\infty$, respectively. For any integer $l \geq -1$, $C_\pm^{(l)}(x)$ vanish rapidly as $x \to \pm\infty$, respectively. The coefficients $C_\pm^{(-1)}(x)$ and $C_\pm^{(0)}(x)$ have the form

$$C_\pm^{(-1)}(x) = \alpha_\pm(x) \begin{pmatrix} 0 & -\mathrm{i} & 0 & -\mathrm{i} \\ -1 & 0 & -1 & 0 \\ 0 & \mathrm{i} & 0 & \mathrm{i} \\ 1 & 0 & 1 & 0 \end{pmatrix} + \beta_\pm(x) \begin{pmatrix} -\mathrm{i} & 0 & -\mathrm{i} & 0 \\ 0 & -1 & 0 & -1 \\ \mathrm{i} & 0 & \mathrm{i} & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \tag{2.23}$$

$$\begin{aligned} C_\pm^{(0)}(x) =& b_{2\pm}(x) \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} + b_{1\pm}(x) \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} + a_{2\pm}(x) \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} + a_{1\pm}(x) \begin{pmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 1 & 0 & -1 & 0 \end{pmatrix} - I, \end{aligned} \tag{2.24}$$

where $\alpha_\pm(x)$, $\beta_\pm(x)$, $a_{2\pm}(x)$, $b_{2\pm}(x)$, and $b_{1\pm}(x)$ are real-valued functions of $x \in \mathbb{R}$, and $a_{1\pm}(x)$ are pure imaginary functions. When $x \to +\infty$, $\alpha_+(x)$, $\beta_+(x)$, $a_{2+}(x)$, and $a_{1+}(x)$ decay to 0 rapidly, while $b_{2+}(x)$ and $b_{1+}(x)$ converge rapidly to $\frac{1}{2}$. Correspondingly, when $x \to -\infty$, $\alpha_-(x)$, $\beta_-(x)$, $a_{2-}(x)$, and $a_{1-}(x)$ decay to 0 rapidly, while $b_{2-}(x)$ and $b_{1-}(x)$ converge rapidly to $\frac{1}{2}$.

Proof. Let us consider the function $\chi(x,k) := P(k)\mu_+(x,k)$. It is easy to check that $\chi(x,k)$ satisfies the integral equation

$$\chi(x,k) = P(k) - \int_x^{+\infty} P(k)\mathrm{e}^{(x-y)\hat{L}(k)}(P^{-1}(k)U_1(y,k)\chi(y,k))dy, \tag{2.25}$$

where

$$U_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ u_0 & v_0 & 0 & 0 \\ v_0 & u_0 & 0 & 0 \end{pmatrix} \tag{2.26}$$

is independent of k. For simplicity, we define $\mathcal{P}(x,y,k) = P(k)\mathrm{e}^{(x-y)L(k)}P^{-1}(k)$. Through direct calculation, we find that $\mathcal{P}(x,y,k)$ is analytic at the origin. Then by analyzing the Voterra integral equation (2.25), we know that $\chi(x,k)$ is analytic in $(-\mathrm{i}S,-S,\mathrm{i}S,S)$, and that $\chi(x,k)$ and all its k-derivative can be continuously extended to $(-\mathrm{i}\bar{S},-\bar{S},\mathrm{i}\bar{S},\bar{S})$. Therefore, $\chi(x,k)$ has the following asymptotic expansion near the origin:

$$\chi(x,k) = \chi(x,0) + \partial_k\chi(x,0)k + \cdots, \quad k \to 0, \quad k \in (-\mathrm{i}\bar{S},-\bar{S},\mathrm{i}\bar{S},\bar{S}), \tag{2.27}$$

where the coefficients are smooth functions of $x \in \mathbb{R}$.

On the other hand, it is easy to see that $P^{-1}(k)$ has a simple pole at $k = 0$ and can be written as

$$P^{-1}(k) = \frac{P^{(-1)}}{k} + P^{(0)}, \tag{2.28}$$

where

$$P^{(-1)} = \begin{pmatrix} 0 & 0 & 0 & -\frac{\mathrm{i}}{2} \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{\mathrm{i}}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad P^{(0)} = \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \end{pmatrix}. \tag{2.29}$$

Therefore, $\mu_+(x,k) = P^{-1}(k)\chi(x,k)$ has at most a simple pole at the origin. Then, we infer that $\mu_+(x,k)$ has the asymptotic expansion of the following form:

$$\mu_+(x,k) = \frac{C_+^{(-1)}(x)}{k} + (I + C^{(0)}(x)) + C_+^{(1)}(x)k + \cdots, \quad k \to 0, \quad k \in (-\mathrm{i}S,-S,\mathrm{i}S,S) \tag{2.30}$$

with

$$C_+^{(-1)}(x) = P^{(-1)}\chi(x,0), \qquad C_+^{(0)}(x) = P^{(-1)}\partial_k\chi(x,0) + P^{(0)}\chi(x,0) - I. \tag{2.31}$$

Using that $\partial_k^j\chi(x,k) \to \partial_k^j P(k)$ rapidly as $x \to +\infty$ and equating powers of k in the identity

$$I = P^{-1}P = \left(\frac{P^{(-1)}}{k} + P^{(0)}\right)(P(0) + P'(0)k),$$

we obtain that $C_+^{(l)}(x)$, $l \geq -1$, vanish rapidly as $x \to +\infty$. Now let us derive the exact form of $C_+^{(l)}(x)$. According to (2.31), there exist functions $a(x)$, $b(x)$, $c(x)$, $d(x)$, $e(x)$, $f(x)$, $g(x)$, and $h(x)$ such that

$$C_+^{(-1)}(x) = \begin{pmatrix} -\frac{\mathrm{i}}{2}e(x) & -\frac{\mathrm{i}}{2}f(x) & -\frac{\mathrm{i}}{2}g(x) & -\frac{\mathrm{i}}{2}h(x) \\ -\frac{1}{2}a(x) & -\frac{1}{2}b(x) & -\frac{1}{2}c(x) & -\frac{1}{2}d(x) \\ \frac{\mathrm{i}}{2}e(x) & \frac{\mathrm{i}}{2}f(x) & \frac{\mathrm{i}}{2}g(x) & \frac{\mathrm{i}}{2}h(x) \\ \frac{1}{2}a(x) & \frac{1}{2}b(x) & \frac{1}{2}c(x) & \frac{1}{2}d(x) \end{pmatrix}. \tag{2.32}$$

Combining symmetries $\mu_+(x,k) = A\mu_+(x,\mathrm{i}k)A^{-1}$ and $\mu_+(x,k) = B\mu_+^*(x,k^*)B$, we get

$$a(x) = c(x) = h(x) = f(x), \qquad b(x) = d(x) = g(x) = e(x), \qquad a(x) = a^*(x), \qquad b(x) = b^*(x).$$

Since $C_+^{(-1)}(x)$ decays rapidly to 0 as $x \to +\infty$, so do $a(x)$ and $b(x)$. Let $\alpha_+(x) = \frac{1}{2}a(x)$, $\beta_+(x) = \frac{1}{2}b(x)$, we complete the proof of (2.23). The proofs for $C_+^{(0)}$ and for $\mu_-(x,k)$ are similar. □

2.3 | Scattering Matrix $s(k)$

When u_0, v_0 have compact support, $\mu_\pm(x,k)$ are analytic for $k \in \mathbb{C} \setminus \{0\}$. Both of them are fundamental solutions of (2.13), so there exists a matrix $s(k)$ independent of x such that $\mu_+(x,k) = \mu_-(x,k)\mathrm{e}^{x\hat{L}(k)}(s(k))$ holds for $k \in \mathbb{C} \setminus \{0\}$. If u_0, v_0 are not compactly supported, then $\mu_+(x,k)$ and $\mu_-(x,k)$ are not defined on the same subset of the complex plane. So we have to define the scattering matrix $s(k)$ in another way. In fact, we can define $s(k)$ by the integral equation

$$s(k) = I - \int_{\mathbb{R}} \mathrm{e}^{-x\hat{L}(k)}(U_2\mu_+)(x,k)dx. \tag{2.33}$$

By analyzing this integral equation, we obtain that $s(k)$ has the following properties.

Proposition 2.4. *Suppose $u_0, v_0 \in \mathcal{S}(\mathbb{R})$, then $s(k)$ defined by (2.33) satisfies the following:*

a. *The domain of matrix $s(k)$ is*

$$\begin{pmatrix} -\mathrm{i}\bar{S} & \mathbb{R}_+e^{\frac{\pi}{4}\mathrm{i}} & \emptyset & \mathbb{R}_+e^{\frac{3\pi}{4}\mathrm{i}} \\ \mathbb{R}_+e^{\frac{\pi}{4}\mathrm{i}} & -\bar{S} & \mathbb{R}_+e^{-\frac{\pi}{4}\mathrm{i}} & \emptyset \\ \emptyset & \mathbb{R}_+e^{-\frac{\pi}{4}\mathrm{i}} & \mathrm{i}\bar{S} & -\mathbb{R}_+e^{\frac{\pi}{4}\mathrm{i}} \\ \mathbb{R}_+e^{\frac{3\pi}{4}\mathrm{i}} & \emptyset & -\mathbb{R}_+e^{\frac{\pi}{4}\mathrm{i}} & \bar{S} \end{pmatrix} \setminus \{0\}, \tag{2.34}$$

   where $\emptyset$ denotes the domain of the element at this position is an empty set.

b. *The matrix $s(k)$ is continuous for k in (2.34), and the diagonal elements are analytic in the interior of their domain.*

c. *For any positive integer j, $\partial_k^j s(k)$ is well-defined and continuous for k in (2.34).*

d. *$s(k)$ has the following symmetries:*

$$s(k) = As(\mathrm{i}k)A^{-1}, \qquad s(k) = Bs^*(k^*)B, \tag{2.35}$$

   where A and B are given by (2.12).

e. *There are diagonal matrices $\{s_j\}_1^\infty$ such that*

$$\left|\partial_k^j\left(s(k) - I - \sum_{l=1}^{N}\frac{s_l}{k^l}\right)\right| = O(k^{-N-1}), \quad k \to \infty \tag{2.36}$$

   uniformly for k in (2.34) and $j = 0, 1, \dots, N$. In particular, when $k \to \infty$, the off-diagonal elements of $s(k)$ vanish rapidly.

f. *For k in* (2.34)*, we have*

$$s(k) = \frac{s^{(-1)}}{k} + O(1), \quad k \to 0, \qquad s^{(-1)} = \begin{pmatrix} \mathfrak{s}_1 & \mathfrak{s}_2 & \emptyset & -\mathfrak{s}_2^* \\ \mathrm{i}\mathfrak{s}_2^* & -\mathrm{i}\mathfrak{s}_1 & -\mathrm{i}\mathfrak{s}_2 & \emptyset \\ \emptyset & \mathfrak{s}_2^* & -\mathfrak{s}_1 & -\mathfrak{s}_2 \\ \mathrm{i}\mathfrak{s}_2 & \emptyset & -\mathrm{i}\mathfrak{s}_2^* & \mathrm{i}\mathfrak{s}_1 \end{pmatrix}, \tag{2.37}$$

   where $\mathfrak{s}_1$ is a pure imaginary number (it can be 0), while $\mathfrak{s}_2$ is uncertain.

g. *If u_0 and v_0 are compactly supported, then $s(k)$ is analytic in $\mathbb{C}\setminus\{0\}$ and satisfy $\mu_+(x,k)=\mu_-(x,k)e^{x\hat{L}(k)}(s(k))$. Moreover, we have* $\det(s(k))=1$.

Proof. See [22, Theorem 3.5] for a similar proof in the context of "good" Boussinesq equation. □

Remark 1. We want to remind the reader that in (2.37), the symbol ∅ indicates that the element at this position is not defined.

2.4 | The Adjoint Spectral Problem

Consider the adjoint Lax pair of Lax pair (2.4)

$$\begin{cases}\phi_x^A=-F^\top\phi^A,\\ \phi_t^A=-G^\top\phi^A,\end{cases}\tag{2.38}$$

where F and G are given by (2.5) and (2.6). We fix $t=0$, and perform spectral analysis on the adjoint Lax pair, then we can get the properties of the adjoint scattering quantities. Next let us describe these results more accurately.

First, we define the adjoint Jost solutions $\mu_\pm^A(x,k)$ as the solutions of the following integral equations:

$$\mu_\pm^A(x,k)=I-\int_{\pm\infty}^{x}\mathrm{e}^{-(x-y)\hat{L}(k)}\left(U_2^\top\mu_\pm^A\right)(y,k)dy.\tag{2.39}$$

By direct calculation, we find that if $\phi(x,k)$ is invertible and satisfies Lax pair (2.4), then $(\phi^{-1})^\top$ satisfies (2.38). So we are not surprised that $\mu_\pm^A(x,k)=(\mu_\pm^{-1})^\top(x,k)$. The following proposition is completely similar to Proposition 2.1, which characterizes the properties of the adjoint Jost solutions.

Proposition 2.5. *Suppose $u_0,v_0\in\mathcal{S}(\mathbb{R})$, then adjoint Jost solutions $\mu_\pm^A(x,k)$ defined by (2.39) has the following properties:*

a. *For each $x\in\mathbb{R}$, the function $\mu_+^A(x,k)$ is well-defined and continuous for $k\in(\mathrm{i}\bar{S},\bar{S},-\mathrm{i}\bar{S},-\bar{S})\setminus\{0\}$. In addition, $\mu_+^A(x,k)$ is analytic for $k\in(\mathrm{i}S,S,-\mathrm{i}S,-S)\setminus\{0\}$.*
b. *For each $x\in\mathbb{R}$, the function $\mu_-^A(x,k)$ is well-defined and continuous for $k\in(-\mathrm{i}\bar{S},-\bar{S},\mathrm{i}\bar{S},\bar{S})\setminus\{0\}$. In addition, $\mu_-^A(x,k)$ is analytic for $k\in(-\mathrm{i}S,-S,\mathrm{i}S,S)\setminus\{0\}$.*
c. *For each positive integer j, the partial derivative $\frac{\partial^j\mu_+^A(x,k)}{\partial k^j}$ of $\mu_+^A(x,k)$ is well-defined and analytic for $k\in(\mathrm{i}S,S,-\mathrm{i}S,-S)\setminus\{0\}$, and it can be continuously extended to the boundary of its analytic region except the origin.*
d. *For each positive integer j, the partial derivative $\frac{\partial^j\mu_-^A(x,k)}{\partial k^j}$ of $\mu_-^A(x,k)$ is well-defined and analytic for $k\in(-\mathrm{i}S,-S,\mathrm{i}S,S)\setminus\{0\}$, and it can be continuously extended to the boundary of its analytic region except the origin.*
e. *For any $n\geq1$ and $\varepsilon>0$, there exist bounded smooth positive functions $\tilde{f}_\pm(x)$ of $x\in\mathbb{R}$ with rapid decay as $x\to\pm\infty$, such that the following estimates hold:*

$$\left|\frac{\partial^j}{\partial k^j}\left(\mu_+^A(x,k)-I\right)\right|\leq\tilde{f}_+(x),\quad k\in(\mathrm{i}\bar{S},\bar{S},-\mathrm{i}\bar{S},-\bar{S}),\quad |k|>\varepsilon,$$

$$\left|\frac{\partial^j}{\partial k^j}\left(\mu_-^A(x,k)-I\right)\right|\leq\tilde{f}_-(x),\quad k\in(-\mathrm{i}\bar{S},-\bar{S},\mathrm{i}\bar{S},\bar{S}),\quad |k|>\varepsilon.$$

f. *$\mu_\pm^A(x,k)$ satisfy the following two symmetries:*

$$\mu_\pm^A(x,k)=A\mu_\pm^A(x,\mathrm{i}k)A^{-1},\quad \mu_\pm^A(x,k)=B\mu_\pm^{A*}(x,k^*)B,\tag{2.40}$$

where A, B are given by (2.12).

g. *When u_0 and v_0 has compact support, then $\mu_\pm^A(x,k)$ are analytic for $k\in\mathbb{C}\setminus\{0\}$ and we have* $\det\mu_\pm^A(x,k)=1$.

h. There are the following relations between $\mu_\pm(x,k)$ and $\mu_\pm^A(x,k)$:

$$J\mu_\pm(x,-k)J^{-1}=\mu_\pm^A(x,k), \tag{2.41}$$

where $J=\operatorname{diag}(\mathrm{i},-1,-\mathrm{i},1)$.

Proof. We only prove the assertion (h). Note that $U_2(x,k)$ satisfies the symmetry $JU_2(x,k)J^{-1}=U_2(x,k)^\top$, and $\mu_\pm(x,k)$ satisfy the integral equation (2.15). It is not difficult to verify that $J\mu_\pm(x,-k)J^{-1}$ satisfy the integral equation (2.39). By uniqueness, we immediately get (2.41). □

Similarly, we need to study the asymptotic behavior of $\mu_\pm^A(x,k)$ when k tends to infinity and tends to 0. Because of the symmetry (2.41), we arrive at the following two propositions without much effort.

Proposition 2.6. *Suppose $u_0,v_0\in S(\mathbb{R})$. Let $p\geq 1$ be an integer, and define $X^{(p)}=I+\sum_{l=1}^p\frac{\mu_+^{[l]}}{k^l}$, $Y^{(p)}=I+\sum_{l=1}^p\frac{\mu_-^{[l]}}{k^l}$. Then there exist smooth positive functions $\tilde g_\pm(x)$ with rapid decay as $x\to\pm\infty$ such that for each integer $j\geq 0$ the estimates*

$$\left|\frac{\partial^j}{\partial k^j}\left(\mu_+^A-X_A^{(p)}\right)\right|\leq\frac{\tilde g(x)}{|k|^{p+1}},\quad x\in\mathbb{R},\quad k\in(\mathrm{i}\bar S,\bar S,-\mathrm{i}\bar S,-\bar S),\quad |k|\geq 2 \tag{2.42a}$$

and

$$\left|\frac{\partial^j}{\partial k^j}\left(\mu_-^A-Y_A^{(p)}\right)\right|\leq\frac{\tilde g(x)}{|k|^{p+1}},\quad x\in\mathbb{R},\quad k\in(-\mathrm{i}\bar S,-\bar S,\mathrm{i}\bar S,\bar S),\quad |k|\geq 2 \tag{2.42b}$$

hold, where $X_A^{(p)}(x,k):=JX^{(p)}(x,-k)J^{-1}$ and $Y_A^{(p)}(x,k):=JY^{(p)}(x,-k)J^{-1}$.

Proposition 2.7. *Suppose $u_0,v_0\in S(\mathbb{R})$, and let $p\geq 0$ be an integer, then there are fourth-order matrix-valued functions $D_\pm^{(l)}(x)$, $l=-1,0,1,\dots,p$, with the following properties: The estimates*

$$\left|\mu_+^A-I-\sum_{l=-1}^p D_+^{(l)}k^l\right|\leq\hat g_+(x)|k|^{p+1-j},\quad x\in\mathbb{R},\quad k\in(\mathrm{i}\bar S,\bar S,-\mathrm{i}\bar S,-\bar S),\quad |k|\leq\frac12 \tag{2.43a}$$

and

$$\left|\mu_-^A-I-\sum_{l=-1}^p D_-^{(l)}k^l\right|\leq\hat g_-(x)|k|^{p+1-j},\quad x\in\mathbb{R},\quad k\in(-\mathrm{i}\bar S,-\bar S,\mathrm{i}\bar S,\bar S),\quad |k|\leq\frac12 \tag{2.43b}$$

hold for each integer $j\geq 0$, where $\hat g_+(x)$ and $\hat g_-(x)$ are bounded smooth positive functions of $x\in\mathbb{R}$ with rapid decay as $x\to\pm\infty$, respectively. For any integer $l\geq -1$, $D_\pm^{(l)}(x)$ vanish rapidly as $x\to\pm\infty$, respectively. The coefficients $D_\pm^{(-1)}(x)$ and $D_\pm^{(0)}(x)$ have the form

$$D_\pm^{(-1)}(x)=-\alpha_\pm(x)\begin{pmatrix}0&-1&0&1\\-\mathrm{i}&0&\mathrm{i}&0\\0&-1&0&1\\-\mathrm{i}&0&\mathrm{i}&0\end{pmatrix}-\beta_\pm(x)\begin{pmatrix}-\mathrm{i}&0&\mathrm{i}&0\\0&-1&0&1\\-\mathrm{i}&0&\mathrm{i}&0\\0&-1&0&1\end{pmatrix},\tag{2.44}$$

$$D_\pm^{(0)}(x)=b_{2\pm}(x)\begin{pmatrix}1&0&-1&0\\0&1&0&-1\\-1&0&1&0\\0&-1&0&1\end{pmatrix}+b_{1\pm}(x)\begin{pmatrix}1&0&1&0\\0&1&0&1\\1&0&1&0\\0&1&0&1\end{pmatrix}$$

$$+a_{2\pm}(x)\begin{pmatrix}0&-\mathrm{i}&0&\mathrm{i}\\ \mathrm{i}&0&-\mathrm{i}&0\\ 0&\mathrm{i}&0&-\mathrm{i}\\ -\mathrm{i}&0&\mathrm{i}&0\end{pmatrix}+a_{1\pm}(x)\begin{pmatrix}0&-\mathrm{i}&0&-\mathrm{i}\\ -\mathrm{i}&0&-\mathrm{i}&0\\ 0&-\mathrm{i}&0&-\mathrm{i}\\ -\mathrm{i}&0&-\mathrm{i}&0\end{pmatrix}-I, \tag{2.45}$$

where $\alpha_\pm(x)$, $\beta_\pm(x)$, $a_{2\pm}(x)$, $b_{2\pm}(x)$, $b_{1\pm}(x)$, and $a_{1\pm}(x)$ are consistent with Proposition 2.3.

At the end of this subsection, we define the adjoint scattering matrix $s^A(k)$ as

$$s^A(k)=I+\int_{-\infty}^{+\infty}\mathrm{e}^{x\hat{L}(k)}\left(U_2^\top\mu_+^A\right)(x,k)dx. \tag{2.46}$$

Then we have the following proposition.

Proposition 2.8. *Suppose $u_0, v_0 \in \mathcal{S}(\mathbb{R})$, then $s^A(k)$ defined by (2.46) satisfies the following:*

a. *The domain of matrix $s^A(k)$ is*

$$\begin{pmatrix}\mathrm{i}\bar{S}&\mathbb{R}_+\mathrm{e}^{\frac{-3\pi}{4}\mathrm{i}}&\emptyset&\mathbb{R}_+\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}\\ \mathbb{R}_+\mathrm{e}^{-\frac{3\pi}{4}\mathrm{i}}&\bar{S}&\mathbb{R}_+\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}}&\emptyset\\ \emptyset&\mathbb{R}_+\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}}&-\mathrm{i}\bar{S}&\mathbb{R}_+\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}\\ \mathbb{R}_+\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}&\emptyset&\mathbb{R}_+\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}&-\bar{S}\end{pmatrix}\setminus\{0\}, \tag{2.47}$$

   where $\emptyset$ denotes the domain of the element at this position is an empty set.

b. *The matrix $s^A(k)$ is continuous for k in (2.47), and the diagonal elements are analytic in the interior of their domain.*

c. *For any positive integer j, $\partial_k^j s^A(k)$ is well-defined and continuous for k in (2.47).*

d. *$s(k)$ has the following symmetries:*

$$s^A(k)=As^A(\mathrm{i}k)A^{-1},\qquad s^A(k)=Bs^{A*}(k^*)B, \tag{2.48}$$

   where A and B are given by (2.12).

e. *There is the following relation between $s(k)$ and $s^A(k)$:*

$$Js(-k)J^{-1}=s^A(k), \tag{2.49}$$

   where $J=\mathrm{diag}(\mathrm{i},-1,-\mathrm{i},1)$.

f. *There are diagonal matrices $\{s_j^A\}_1^\infty$ such that*

$$\left|\partial_k^j\left(s^A(k)-I-\sum_{l=1}^{N}\frac{s_l^A}{k^l}\right)\right|=O(k^{-N-1}),\quad k\to\infty \tag{2.50}$$

   uniformly for k in (2.47) and $j=0,1,\dots,N$. In particular, when $k\to\infty$, the off-diagonal elements of $s^A(k)$ vanish rapidly.

g. *For k in* (2.47)*, we have*

$$s^A(k)=\frac{s^{A(-1)}}{k}+O(1),\quad k\to 0, \tag{2.51}$$

   where $s^{A(-1)}=-Js^{(-1)}J^{-1}$.

h. *If u_0 and v_0 are compactly supported, then $s^A(k)$ is analytic in $\mathbb{C}\setminus\{0\}$ and satisfies $\mu_+^A(x,k)=\mu_-^A(x,k)\mathrm{e}^{-x\hat{L}(k)}(s^A(k))$. Moreover, we have $\det s^A(k)=1$.*

Remark 2. The proof for assertion (e) is completely similar to the proof for assertion (h) in Proposition 2.5. The assertions (f) and (g) are the results of combining (2.36), (2.37), and (2.49).

2.5 | Fundamental Tensors

In this subsection, we introduce some results in Ref. [16], which will be used in Section 2.6. To facilitate the following analysis, we arbitrarily fix an integer j from $\{1, 2, \dots, 7, 8\}$. Suppose $k \in D_j$, we now rearrange the order of $l_1(k), \dots, l_4(k)$ such that

$$\mathrm{Re}(l_{j_1}) > \mathrm{Re}(l_{j_2}) > \mathrm{Re}(l_{j_3}) > \mathrm{Re}(l_{j_4}). \tag{2.52}$$

Obviously, this order depends on the position of k, that is, when k is in different D_j, the order of $l_i(k), i = 1, \dots, 4$ is also different.

Definition 2.1. For $n = 1, \dots, 4$, we define the totally antisymmetric fundamental tensors $f_n(x,k)$ and $g_n(x,k)$ for $k \in D_j$:

$$\begin{aligned} f_n(x,k) &= \mu_{-j_1}(x,k) \wedge \mu_{-j_2}(x,k) \wedge \dots \wedge \mu_{-j_n}(x,k), \\ g_n(x,k) &= \mu_{+j_n}(x,k) \wedge \mu_{+j_{n+1}}(x,k) \wedge \dots \wedge \mu_{+j_4}(x,k), \end{aligned} \tag{2.53}$$

where $\mu_{\pm j_i}(x,k)$ is the j_ith column of $\mu_\pm(x,k)$ and $\{j_1, \dots, j_n\}$ is consistent with (2.52).

The following lemma is proved in [16].

Lemma 2.2. *Let $1 \le n \le 4$ be an integer. Assume that $\mathbf{A}$ is a linear transformation on $\mathbb{C}^4$. Then there exists a unique linear transformation $\mathbf{A}^{(n)} : \wedge^n(\mathbb{C}^4) \to \wedge^n(\mathbb{C}^4)$ such that for all $u_1, \dots, u_n \in \mathbb{C}^4$,*

$$\mathbf{A}^{(n)}(u_1 \wedge \dots \wedge u_n) = \sum_{j=1}^{n} u_1 \wedge \dots \wedge u_{j-1} \wedge \mathbf{A}u_j \wedge u_{j+1} \wedge \dots \wedge u_n. \tag{2.54}$$

Recall that $\mu_{\pm j_i}(x,k)$ satisfy equation

$$\partial_x \mu_{\pm j_i}(x,k) = (kJ - l_{j_i} I + U_2(x,k))\mu_{\pm j_i}(x,k), \tag{2.55}$$

then $f_n(x,k)$ and $g_n(x,k)$ must satisfy the extended differential equations

$$\begin{cases} \partial_x f_n(x,k) = (kJ^{(n)} - (l_{j_1} + \dots + l_{j_n})I + U_2^{(n)}(x,k))f_n(x,k), \\ \lim_{x\to-\infty} f_n(x,k) = e_{j_1} \wedge \dots \wedge e_{j_n}, \end{cases} \tag{2.56}$$

and

$$\begin{cases} \partial_x g_n(x,k) = (kJ^{(5-n)} - (l_{j_n} + \dots + l_{j_4})I + U_2^{(5-n)}(x,k))g_n(x,k), \\ \lim_{x\to+\infty} g_n(x,k) = e_{j_n} \wedge \dots \wedge e_{j_4}, \end{cases} \tag{2.57}$$

respectively, where $J^{(n)}$ and $U_2^{(n)}(x,k)$ denote the nth order extensions (2.54) of J and $U_2(x,k)$ to $\wedge^n(\mathbb{C}^4)$, and $\{e_1, \dots, e_4\}$ represents the standard basis in $\mathbb{C}^4$.

The following proposition gives the analytic properties of $f_n(x,k)$ and $g_n(x,k)$.

Proposition 2.9. *Suppose $u_0, v_0 \in S(\mathbb{R})$. For each $k \in D_j$, we can define a family of fundamental tensors $f_n(x,k)$ and $g_n(x,k)$ according to (2.53) for $n = 1, 2, 3, 4$. Moreover, for each integer $l \ge 0$, $\frac{\partial^l f_n}{\partial k^l}(x,k)$ and $\frac{\partial^l g_n}{\partial k^l}(x,k)$ are analytic in D_j and can be continuously extended to $\bar{D}_j \setminus \{0\}$.*

Proof. $f_n(x,k)$ and $g_n(x,k)$ are obviously well-defined by (2.53), we prove their analytic properties below. First, for a given integer n, $1 \le n \le 4$, we define the normal operators appearing in differential equations (2.56) and (2.57) by

$$A_n = kJ^{(n)} - (l_{j_1} + \dots + l_{j_n})I, \qquad B_n = kJ^{(5-n)} - (l_{j_n} + \dots + l_{j_4})I. \tag{2.58}$$

It is not difficult to verify that the standard basis tensors

$$\left\{ e_{i_1} \wedge e_{i_2} \ldots \wedge e_{i_n} \mid 1 \le i_1 < \ldots < i_n \le 4 \right\}$$

of $\wedge^n(\mathbb{C}^4)$ are exactly the eigenvectors of operators A_n and B_n. Then, the spectrum of operators A_n and B_n is

$$\operatorname{spec}(A_n) = \left\{ (l_{i_1} + l_{i_2} + \cdots + l_{i_n}) - (l_{j_1} + l_{j_2} + \cdots + l_{j_n}) \mid 1 \le i_1 < \ldots < i_n \le 4 \right\},$$

$$\operatorname{spec}(B_n) = \left\{ (l_{i_n} + l_{i_{n+1}} + \cdots + l_{i_4}) - (l_{j_n} + l_{j_{n+1}} + \cdots + l_{j_4}) \mid 1 \le i_n < \ldots < i_4 \le 4 \right\}.$$

Therefore, the real part of the spectrum of operator A_n is always nonpositive, and the real part of the spectrum of operator B_n is always nonnegative.

As before, we hope to analyze the integra equations satisfied by $f_n(x,k)$ and $g_n(x,k)$ to obtain their analytical properties. Note that $f_n(x,k)$ and $g_n(x,k)$ satisfy differential equations (2.56) and (2.57). For convenience, we reduce these eight differential equations to an abstract model. Let V be a finite-dimensional Hermitian vector space with norm $\|\cdot\|$, $\mathbf{A}$ be a normal operator on the vector space, playing the role of A_n. Therefore, we assume that the real part of the spectrum of operator $\mathbf{A}$ is always nonpositive. And let $\mathbf{q}(x,k)$ is a linear operator of the form

$$\mathbf{q}(x,k) = \frac{\mathbf{q}_1(x)}{k}, \qquad \mathbf{q}_1 \in \mathcal{S}(\mathbb{R}; \mathcal{L}(V)) \tag{2.59}$$

on V, playing the role of $U_2^{(n)}$. Then, for fixed $k \in D_j$ and fixed $\mathbf{u}_0 \in \ker\mathbf{A}$, we consider the following "model problem":

$$\partial_x \mathbf{u} = \mathbf{A}(k)\mathbf{u} + \mathbf{q}(x,k)\mathbf{u}, \qquad \lim_{x \to -\infty} \mathbf{u}(x) = \mathbf{u}_0. \tag{2.60}$$

The problem for $g_n(x,k)$ can be brought to this form by simply changing x to $-x$. Therefore, we only need to analyze the model problem (2.60). For any real s and t, we deduce from (2.60) that $\mathbf{u}$ must satisfy the integral equation

$$\mathbf{u}(x) = \mathrm{e}^{(x-s)\mathbf{A}(k)}\mathbf{u}(s) + \int_s^x \mathrm{e}^{(x-y)\mathbf{A}(k)}\mathbf{q}(y,k)\mathbf{u}(y)dy. \tag{2.61}$$

Note that the real part of the normal operator $\mathbf{A}$ is nonpositive, so the norm of $\mathrm{e}^{t\mathbf{A}}$ is ≤ 1 when $t \ge 0$. By taking $s \to -\infty$ and applying the boundary conditions in (2.60), we can get the desired Volterra integral equation

$$\mathbf{u}(x) = \mathbf{u}_0 + \int_{-\infty}^x \mathrm{e}^{(x-y)\mathbf{A}(k)}\mathbf{q}(y,k)\mathbf{u}(y)dy. \tag{2.62}$$

The Picard iteration series associated with (2.62) is defined by

$$\begin{cases} \mathbf{u}^{(0)}(x) \equiv \mathbf{u}_0, \\ \mathbf{u}^{(\nu+1)}(x) = \mathbf{u}_0 + \displaystyle\int_{-\infty}^x \mathrm{e}^{(x-y)\mathbf{A}(k)}\mathbf{q}(y,k)\mathbf{u}^{(\nu)}(y)dy. \end{cases} \tag{2.63}$$

Next, we only need to follow the proof steps of [16, Theorem 3.7] to analyze the Picard iterates series, and then we can get the conclusion of the proposition. □

Remark 3. In fact, Proposition 2.9 contains the assertions (a)–(d) in Proposition 2.1.

As a direct result of Proposition 2.9, we have

Corollary 2.1. *Assume $u_0, v_0 \in \mathcal{S}(\mathbb{R})$. Then,*

- *$\mu_{+1}(x,k) \wedge \mu_{+2}(x,k)$ and $\mu_{-3}(x,k) \wedge \mu_{-4}(x,k)$ are well-defined and continuous for $k \in \bar{D}_1 \cup \bar{D}_2 \setminus \{0\}$. In addition, they are analytic in the interior of $\bar{D}_1 \cup \bar{D}_2$.*
- *$\mu_{+2}(x,k) \wedge \mu_{+3}(x,k)$ and $\mu_{-1}(x,k) \wedge \mu_{-4}(x,k)$ are well-defined and continuous for $k \in \bar{D}_7 \cup \bar{D}_8 \setminus \{0\}$. In addition, they are analytic in the interior of $\bar{D}_7 \cup \bar{D}_8$.*

Lemma 2.3. *Let* $\mathbf{A}$ *be a linear transformation on* $\mathbb{C}^N$*, and* $\mathbf{A}^{(N)}$ *denote the N-order extension (2.54) of* $\mathbf{A}$ *on* $\wedge^N(\mathbb{C}^N)$*. Then for any* $C \in \wedge^N(\mathbb{C}^N)$*, we have*

$$\mathbf{A}^{(N)}(C) = \mathrm{trace}(\mathbf{A})C,$$

where $\mathrm{trace}(\mathbf{A})$ *represents the trace of matrix* $\mathbf{A}$*.*

Proof. Suppose that C can be written as $C_1 \wedge C_2 \wedge \cdots \wedge C_{N-1} \wedge C_N$, where $C_j \in \mathbb{C}^N$, $j = 1, 2, \ldots, N$. Then obviously we have

$$\begin{aligned}
\mathbf{A}^{(N)}(C) &= \frac{d((I+t\mathbf{A})C_1 \wedge (I+t\mathbf{A})C_2 \wedge \cdots (I+t\mathbf{A})C_{N-1} \wedge (I+t\mathbf{A})C_N)}{dt}|_{t=0} \\
&= \frac{d(\det\left((I+t\mathbf{A})C_1, \ldots, (I+t\mathbf{A})C_N\right))}{dt}|_{t=0} e_1 \wedge e_2 \wedge \cdots \wedge e_{N-1} \wedge e_N \\
&= \frac{d(\det(I+t\mathbf{A}))}{dt}|_{t=0} \det\left(C_1, \ldots, C_N\right) e_1 \wedge e_2 \wedge \cdots \wedge e_{N-1} \wedge e_N \\
&= \mathrm{trace}(\mathbf{A})C_1 \wedge C_2 \wedge \cdots \wedge C_{N-1} \wedge C_N.
\end{aligned}$$

□

Define $\phi_{\pm j}(x,k) = \mu_{\pm j}(x,k)\mathrm{e}^{xl_j}$. Now we are ready to introduce some spectral functions, which are very important in our inverse scattering analysis.

Lemma 2.4. *Assume* $u_0, v_0 \in S(\mathbb{R})$*. Then, there exist functions* $\Delta(k)$*,* $\Xi(k)$*,* $\Lambda(k)$ *such that*

$$\mu_{+1}(x,k) \wedge \mu_{+2}(x,k) \wedge \mu_{-3}(x,k) \wedge \mu_{-4}(x,k) = \Delta(k) e_1 \wedge e_2 \wedge e_3 \wedge e_4, \tag{2.64}$$

$$\mu_{+2}(x,k) \wedge \mu_{+3}(x,k) \wedge \mu_{-1}(x,k) \wedge \mu_{-4}(x,k) = \Xi(k) e_1 \wedge e_2 \wedge e_3 \wedge e_4, \tag{2.65}$$

$$\phi_{+2}(x,k) \wedge \phi_{+3}(x,k) \wedge \phi_{-3}(x,k) \wedge \phi_{-4}(x,k) = \Lambda(k) e_1 \wedge e_2 \wedge e_3 \wedge e_4. \tag{2.66}$$

Moreover,

$\Delta(k)$ *is well-defined and continuous for* $k \in \bar{D}_1 \cup \bar{D}_2 \setminus \{0\}$*, and is analytic in the interior of* $\bar{D}_1 \cup \bar{D}_2$*.*

$\Xi(k)$ *is well-defined and continuous for* $k \in \bar{D}_7 \cup \bar{D}_8 \setminus \{0\}$*, and is analytic in the interior of* $\bar{D}_7 \cup \bar{D}_8$*.*

$\Lambda(k)$ *is well-defined and is a smooth function on* $(0, +\infty)$*.*

Proof. Since $e_1 \wedge e_2 \wedge e_3 \wedge e_4$ is the basis of the linear space $\wedge^4(\mathbb{C}^4)$, we obtain the existence of functions $\Delta(k)$, $\Xi(k)$, and $\Lambda(k)$. Obviously, their analytic properties are the direct result of Corollary 2.1. Now it remains to show that these functions are independent of x. In fact, one can easily verify that $\mu_{+1}(x,k) \wedge \mu_{+2}(x,k) \wedge \mu_{-3}(x,k) \wedge \mu_{-4}(x,k)$ and $\mu_{+2}(x,k) \wedge \mu_{+3}(x,k) \wedge \mu_{-1}(x,k) \wedge \mu_{-4}(x,k)$ satisfy the following extended differential equation:

$$\partial_x f(x,k) = U_2^{(4)}(x,k) f(x,k). \tag{2.67}$$

Similarly, $\phi_{+2}(x,k) \wedge \phi_{+3}(x,k) \wedge \phi_{-3}(x,k) \wedge \phi_{-4}(x,k)$ satisfies differential equation

$$\partial_x f(x,k) = F^{(4)}(x,k) f(x,k). \tag{2.68}$$

From (2.5) and (2.7), we know that $\mathrm{trace}(F) = 0$, $\mathrm{trace}(U_2) = 0$. Then as a result of Lemma 2.3, we obtain that the derivative of each tensor with respect to x is 0. So we have completed the proof of the lemma. □

Remark 4. We note that f_4 and g_1 also satisfy (2.67), so we know that neither f_4 nor g_1 depends on x. Moreover, we have $f_4 = g_1 = e_1 \wedge e_2 \wedge e_3 \wedge e_4$ by combining the boundary conditions.

In fact, all the analysis in this section so far can be applied to the adjoint Lax pair (2.38). For the convenience of later use, we give several results of adjoint scattering quantities:

- $\mu_{-1}^{A}(x,k) \wedge \mu_{-2}^{A}(x,k)$ and $\mu_{+3}^{A}(x,k) \wedge \mu_{+4}^{A}(x,k)$ are well-defined and continuous for $k \in \bar{D}_1 \cup \bar{D}_2 \setminus \{0\}$. In addition, they are analytic in the interior of $\bar{D}_1 \cup \bar{D}_2$.
- $\mu_{-2}^{A}(x,k) \wedge \mu_{-3}^{A}(x,k)$ and $\mu_{+1}^{A}(x,k) \wedge \mu_{+4}^{A}(x,k)$ are well-defined and continuous for $k \in \bar{D}_7 \cup \bar{D}_8 \setminus \{0\}$. In addition, they are analytic in the interior of $\bar{D}_7 \cup \bar{D}_8$.
- There exist a function $\mathrm{Y}(k)$ independent of x such that $\mu_{-1}^{A}(x,k) \wedge \mu_{-2}^{A}(x,k) \wedge \mu_{+3}^{A}(x,k) \wedge \mu_{+4}^{A}(x,k) = \mathrm{Y}(k) e_1 \wedge e_2 \wedge e_3 \wedge e_4$. $\mathrm{Y}(k)$ is well-defined and continuous for $k \in \bar{D}_1 \cup \bar{D}_2 \setminus \{0\}$. In addition, it is analytic in the interior of $\bar{D}_1 \cup \bar{D}_2$.

Next, we construct a series of compactly supported initial data $\{u_0^{(i)}, v_0^{(i)}\}$ from the initial data $\{u_0, v_0\}$, and we show the relationship between the scattering quantities associated with $\{u_0, v_0\}$ and the scattering quantities associated with $\{u_0^{(i)}, v_0^{(i)}\}$.

Let $\eta \in C_c^{\infty}(\mathrm{R})$ be a cutoff function which equals one for $|x| \le 1$ and which vanishes for $|x| \ge 2$. For any positive integer i, let $\eta_i(x) = \eta(\frac{x}{i})$. We define

$$u_0^{(i)} = u_0 \cdot \eta_i(x), \quad v_0^{(i)} = v_0 \cdot \eta_i(x). \tag{2.69}$$

Note that $u_0^{(i)}$ and $v_0^{(i)}$ also belong to $\mathcal{S}(\mathbb{R})$, and they are smooth and compactly supported. In the previous section, if we replace u_0 and v_0 with $u_0^{(i)}$ and $v_0^{(i)}$, then all analyses are still valid. We use the superscript (i) to represent the scattering quantities related to $u_0^{(i)}$ and $v_0^{(i)}$. For example, we use $f_n^{(i)}(x,k)$ and $g_n^{(i)}(x,k)$ to represent fundamental tensors associated with $u_0^{(i)}$ and $v_0^{(i)}$.

Remark 5. The superscript (i) here is easily confused with the nth order extensions (2.54) of a matrix. One can distinguish them according to the context.

Lemma 2.5. *Assume $u_0, v_0 \in \mathcal{S}(\mathbb{R})$. Let $u_0^{(i)}$, $v_0^{(i)}$ and $f_n^{(i)}(x,k)$, $g_n^{(i)}(x,k)$ be defined as above, then for $n = 1,2,3,4$, we have*

$$\lim_{i\to\infty} f_n^{(i)}(x,k) = f_n(x,k), \quad x \in \mathbb{R}, \quad k \in \bar{D}_j \setminus \{0\}, \tag{2.70}$$

$$\lim_{i\to\infty} g_n^{(i)}(x,k) = g_n(x,k), \quad x \in \mathbb{R}, \quad k \in \bar{D}_j \setminus \{0\}. \tag{2.71}$$

Proof. First, we define

$$\Lambda_1 = \begin{pmatrix} -\mathrm{i} & 0 & -\mathrm{i} & 0 \\ 0 & -1 & 0 & -1 \\ \mathrm{i} & 0 & \mathrm{i} & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad \Lambda_2 = \begin{pmatrix} 0 & -\mathrm{i} & 0 & -\mathrm{i} \\ -1 & 0 & -1 & 0 \\ 0 & \mathrm{i} & 0 & \mathrm{i} \\ 1 & 0 & 1 & 0 \end{pmatrix},$$

$$\tilde{\Delta}(x,k) = \frac{|u_0|}{2|k|}\left\|\Lambda_1^{(n)}\right\| + \frac{|v_0|}{2|k|}\left\|\Lambda_2^{(n)}\right\|,$$

where $\Lambda_1^{(n)}$ and $\Lambda_2^{(n)}$ denote the nth-order extensions (2.54) of Λ_1 and Λ_2 to $\wedge^n(\mathbb{C}^4)$. It is easy to find $\tilde{\Delta}(x,k) \in L^1(\mathbb{R})$ for each $k \in \bar{D}_j \setminus \{0\}$. Let us replace u and v in matrix U_2 with $u^{(i)}$ and $v^{(i)}$, and the matrix is denoted by $U_2^{(i)}$. The nth-order extension of $U_2^{(i)}$ is defined as $(U_2^{(i)})^{(n)}$. Then, it is easy to see that

$$\left\|U_2^{(n)}(x,k)\right\| \le \tilde{\Delta}(x,k), \qquad \left\|\left(U_2^{(i)}\right)^{(n)}(x,k)\right\| \le \tilde{\Delta}(x,k). \tag{2.72}$$

Recalling (2.62), we find $f_n(x,k)$ and $f_n^{(i)}(x,k)$ satisfy the integral equations

$$f_n(x,k) = \mathbf{u}_0 + \int_{-\infty}^{x} \mathrm{e}^{(x-y)\mathbf{A}(k)} U_2^{(n)}(y,k) f_n(y,k) dy \tag{2.73}$$

and

$$f_n^{(i)}(x,k)=\mathbf{u}_0+\int_{-\infty}^{x}\mathrm{e}^{(x-y)\mathbf{A}(k)}\left(U_2^{(i)}\right)^{(n)}(y,k)f_n^{(i)}(y,k)dy, \tag{2.74}$$

respectively. Here, $\mathbf{u}_0=e_{j_1}\wedge\ldots\wedge e_{j_n}$ and $\mathbf{A}(k)=kJ^{(n)}-(l_{j_1}+\cdots+l_{j_n})I$. Hence for each $x\in\mathbb{R}$, $k\in\bar{D}_j\setminus\{0\}$ we have

$$\begin{aligned}\left\|f_n(x,k)-f_n^{(i)}(x,k)\right\|=&\left\|\int_{-\infty}^{x}\mathrm{e}^{(x-y)\mathbf{A}(k)}(U_2^{(n)}f_n-\left(U_2^{(i)}\right)^{(n)}f_n^{(i)})(y,k)dy\right\|\\ \le&\int_{-\infty}^{x}\left\|U_2^{(n)}(y,k)-\left(U_2^{(i)}\right)^{(n)}(y,k)\right\|\|f_n(y,k)\|dy\\ &+\int_{-\infty}^{x}\left\|f_n(y,k)-f_n^{(i)}(y,k)\right\|\left\|\left(U_2^{(i)}\right)^{(n)}(y,k)\right\|dy.\end{aligned} \tag{2.75}$$

On the other hand, by analyzing the Picard iteration series (2.63), it is easy to verify that $f_n(x,k)$ and $f_n^{(i)}(x,k)$ obey the estimates

$$\begin{aligned}&\|f_n(x,k)\|\le\|\mathbf{u}_0\|\mathrm{e}^{Q(x,k)},\\ &\left\|f_n^{(i)}(x,k)\right\|\le\|\mathbf{u}_0\|\mathrm{e}^{Q^{(i)}(x,k)},\end{aligned} \tag{2.76}$$

where

$$Q(x,k)=\int_{-\infty}^{x}\left\|U_2^{(n)}(y,k)\right\|dy,\quad Q^{(i)}(x,k)=\int_{-\infty}^{x}\left\|\left(U_2^{(i)}\right)^{(n)}(y,k)\right\|dy.$$

By (2.72), we conclude that

$$\|f_n(x,k)\|\le\|\mathbf{u}_0\|\mathrm{e}^{\|\tilde{\Delta}(x,k)\|_{L^1(\mathbb{R})}},\qquad\left\|f_n^{(i)}(x,k)\right\|\le\|\mathbf{u}_0\|\mathrm{e}^{\|\tilde{\Delta}(x,k)\|_{L^1(\mathbb{R})}}.$$

Then it is easy to see that the integrands of the two integrals on the right side of (2.75) are bounded by an integrable function. Let $\xi(x,k)=\overline{\lim}_{i\to\infty}\left\|f_n(x,k)-f_n^{(i)}(x,k)\right\|$, take the upper limit on both sides of (2.75), and apply the inverse Fatou's lemma to obtain

$$\xi(x,k)\le\int_{-\infty}^{x}\left\|U_2^{(n)}(y,k)\right\|\xi(y,k)dy. \tag{2.77}$$

By Gronwall's inequality, we deduce $\xi(x,k)\equiv0$, which implies (2.70). (2.71) can be similarly verified. □

Corollary 2.2. *Assume $u_0,v_0\in S(\mathbb{R})$. Let $\{\mu_\pm(x,k);\mu_\pm^A(x,k)\}$ and $\left\{\mu_\pm^{(i)}(x,k);(\mu_\pm^A)^{(i)}(x,k)\right\}$ be Jost solutions and adjoint Jost solutions associated with u_0,v_0 and $u_0^{(i)},v_0^{(i)}$, respectively. Then, we have*

$$\lim_{i\to\infty}\mu_+^{(i)}(x,k)=\mu_+(x,k),\quad x\in\mathbb{R},\quad k\in(-\mathrm{i}\bar{S},-\bar{S},\mathrm{i}\bar{S},\bar{S})\setminus\{0\}, \tag{2.78}$$

$$\lim_{i\to\infty}\mu_-^{(i)}(x,k)=\mu_-(x,k),\quad x\in\mathbb{R},\quad k\in(\mathrm{i}\bar{S},\bar{S},-\mathrm{i}\bar{S},-\bar{S})\setminus\{0\}, \tag{2.79}$$

$$\lim_{i\to\infty}(\mu_+^A)^{(i)}(x,k)=\mu_+^A(x,k),\quad x\in\mathbb{R},\quad k\in(\mathrm{i}\bar{S},\bar{S},-\mathrm{i}\bar{S},-\bar{S})\setminus\{0\}, \tag{2.80}$$

$$\lim_{i\to\infty}(\mu_-^A)^{(i)}(x,k)=\mu_-^A(x,k),\quad x\in\mathbb{R},\quad k\in(-\mathrm{i}\bar{S},-\bar{S},\mathrm{i}\bar{S},\bar{S})\setminus\{0\}. \tag{2.81}$$

Proof. Equations (2.78) and (2.79) are the direct results of Lemma 2.5, then (2.80) and (2.81) are obtained by combining symmetry (2.41). □

Corollary 2.3. *Let $\{s^{(i)}, s^{A(i)}\}$ denote the scattering matrices associated with potential functions $\{u_0^{(i)}, v_0^{(i)}\}$. Then*

$$\Delta(k) = \lim_{i\to\infty} \det \begin{pmatrix} s_{11}^{(i)} & s_{12}^{(i)} \\ s_{21}^{(i)} & s_{22}^{(i)} \end{pmatrix}, \qquad k \in \bar{D}_1 \cup \bar{D}_2 \setminus \{0\}, \tag{2.82}$$

$$\Xi(k) = \lim_{i\to\infty} \det \begin{pmatrix} s_{22}^{(i)} & s_{23}^{(i)} \\ s_{32}^{(i)} & s_{33}^{(i)} \end{pmatrix}, \qquad k \in \bar{D}_7 \cup \bar{D}_8 \setminus \{0\}, \tag{2.83}$$

$$\Upsilon(k) = \lim_{i\to\infty} \det \begin{pmatrix} s_{33}^{A(i)} & s_{34}^{A(i)} \\ s_{43}^{A(i)} & s_{44}^{A(i)} \end{pmatrix}, \qquad k \in \bar{D}_1 \cup \bar{D}_2 \setminus \{0\}, \tag{2.84}$$

$$\Lambda(k) = \lim_{i\to\infty} \det \begin{pmatrix} s_{12}^{(i)} & s_{13}^{(i)} \\ s_{22}^{(i)} & s_{23}^{(i)} \end{pmatrix}, \qquad k \in \mathbb{R}_+. \tag{2.85}$$

Proof. The result is obtained by using Lemmas 2.4, 2.5 and the formula

$$\mu_+^{(i)}(x,k)\mathrm{e}^{xL(k)} = \mu_-^{(i)}(x,k)\mathrm{e}^{xL(k)}s^{(i)}(k), \quad k \in \mathbb{C} \setminus \{0\}. \qquad \square$$

Remark 6. In fact, one can verify $\Delta(k) = \Upsilon(k)$ by direct calculation.

Now we introduce the definition of generalized cross product [18, 31].

Definition 2.2 (Generalized cross product). For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^4$, we define

$$L(\mathbf{u},\mathbf{v},\mathbf{w}) = \det \begin{pmatrix} u_1 & u_2 & u_3 & u_4 \\ v_1 & v_2 & v_3 & v_4 \\ w_1 & w_2 & w_3 & w_4 \\ e_1 & e_2 & e_3 & e_4 \end{pmatrix}, \tag{2.86}$$

where u_i, v_i, and w_i represent the components of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, respectively, and $\{e_1, \dots, e_4\}$ represents the standard basis in $\mathbb{C}^4$.

Similar to the usual three-dimensional crossed product, $L(\cdot)$ is multilinear and totally antisymmetric. The analytic properties of $L(\mathbf{u},\mathbf{v},\mathbf{w})$ and $\mathbf{u}\wedge\mathbf{v}\wedge\mathbf{w}$ are the same, but the difference is $L(\mathbf{u},\mathbf{v},\mathbf{w}) \in \mathbb{C}^4$, which allows us to use them to express the fundamental meromorphic eigenfunctions $M(x,k)$ (see Lemma 2.11). This observation also leads to the following lemma.

Lemma 2.6. *If $u_0, v_0 \in \mathcal{S}(\mathbb{R})$ and the generalized cross product $L(\cdot)$ is defined by (2.86), then*

- *$L(\mu_{-2}^A, \mu_{+3}^A, \mu_{+4}^A)$, $L(\mu_{-1}^A, \mu_{-2}^A, \mu_{+4}^A)$, $L(\mu_{+2}, \mu_{-3}, \mu_{-4})$, and $L(\mu_{+1}, \mu_{+2}, \mu_{-4})$ are well-defined and continuous for $k \in \bar{D}_1 \setminus \{0\}$, and are analytic in D_1.*
- *$L(\mu_{-2}^A, \mu_{-3}^A, \mu_{+4}^A)$, $L(\mu_{+1}^A, \mu_{-2}^A, \mu_{+4}^A)$, $L(\mu_{+2}, \mu_{+3}, \mu_{-4})$, and $L(\mu_{-1}, \mu_{+2}, \mu_{-4})$ are well-defined and continuous for $k \in \bar{D}_8 \setminus \{0\}$, and are analytic in D_8.*

Next, we give a lemma that will be used later.

Lemma 2.7. *Suppose that $a_1, \dots, a_4 \in \mathbb{C}^4$ and let $A = (a_1, a_2, a_3, a_4)$. If A is invertible, then*

$$(A^{-1})^\top = \frac{1}{\det A}(-L(a_2,a_3,a_4), L(a_1,a_3,a_4), -L(a_1,a_2,a_4), L(a_1,a_2,a_3)). \tag{2.87}$$

2.6 | The Fundamental Meromorphic Eigenfunction $M(x,k)$

In order to formulate an RH problem, we need to define a piecewise meromorphic matrix-valued solution $M(x,k)$ of (2.13). We denote $M(x,k)$ as $M_n(x,k)$ when $k \in D_n$, $1 \le n \le 8$. Then according to [22], we can use the following integral equations to define $M_n(x,k)$:

$$(M_n)_{ij}(x,k) = \delta_{ij} + \int_{\gamma_{ij}^n} (\mathrm{e}^{(x-x')\hat{L}(k)}(U_2 M_n)(x',k))_{ij} dx', \tag{2.88}$$

where

$$\gamma_{ij}^n = \begin{cases} (-\infty, x) := \gamma_1, & \mathrm{Re}l_i(x) < \mathrm{Re}l_j(x), \\ (+\infty, x) := \gamma_2, & \mathrm{Re}l_i(x) \ge \mathrm{Re}l_j(x). \end{cases} \tag{2.89}$$

Let us define $\mathcal{Q} = \{0\} \cup \mathcal{Z}$, where $\mathcal{Z}$ denotes the zero of the Fredholm determinant associated with (2.88), in particular, $\mathcal{Z} \cap \bar{D}_1 = \cup_{j=1}^4 \left\{k \in \bar{D}_1 \mid \tilde{f}_j(k) = 0\right\}$, where $\tilde{f}_j(k)$ are the Fredholm determinants given explicitly in (2.105). The following proposition gives the basic properties of $M(x,k)$.

Proposition 2.10. *Suppose $u_0, v_0 \in S(\mathbb{R})$, then we can define eight matrix-valued functions $M_n(x,k)$ by (2.88), which have the following properties:*

a. *The matrix-valued function $M_n(x,k)$ is well-defined for $x \in \mathbb{R}$ and $k \in \bar{D}_n \setminus \mathcal{Q}$, and for each $k \in \bar{D}_n \setminus \mathcal{Q}$, $M_n(\cdot,k)$ is smooth and satisfies the (2.13).*
b. *For each $x \in \mathbb{R}$, $M_n(x,k)$ is continuous for $k \in \bar{D}_n \setminus \mathcal{Q}$ and analytic in $D_n \setminus \mathcal{Q}$.*
c. *For any $\varepsilon > 0$, there is a constant C depending on ε such that*

$$|M_n(x,k)| \le C, \quad x \in \mathbb{R}, \quad k \in \bar{D}_n, \quad \mathrm{dist}(k, \mathcal{Q}) \ge \varepsilon. \tag{2.90}$$

d. *For each $x \in \mathbb{R}$ and any positive integer j, $\frac{\partial^j M_n(x,k)}{\partial k^j}$ can be continuously extended to $\bar{D}_n \setminus \mathcal{Q}$.*
e. $\det(M_n)(x,k) = 1$ *for $x \in \mathbb{R}$ and $k \in \bar{D}_n \setminus \mathcal{Q}$.*
f. *For each $x \in \mathbb{R}$, $M(x,k)$ satisfies the following two symmetries:*

$$M(x,k) = AM(x,\mathrm{i}k)A^{-1}, \quad M(x,k) = BM^*(x,k^*)B, \tag{2.91}$$

where A, B are given by (2.12).
g. *Define $M^A(x,k)$ as $(M^{-1})^\top(x,k)$ for $x \in \mathbb{R}$ and $k \in \bar{D}_n \setminus \mathcal{Q}$, then we have*

$$JM(x,-k)J^{-1} = M^A(x,k), \quad k \in \bar{D}_n \setminus \mathcal{Q}, \tag{2.92}$$

where $J = \mathrm{diag}(\mathrm{i}, -1, -\mathrm{i}, 1)$.

Proof. We prove this proposition for $n = 1$. Let us consider the fourth column of $M_1(x,k)$. Define function $H(x)$ as follows:

$$H(x) = \begin{cases} 1, & x > 0, \\ 0, & x \le 0. \end{cases}$$

And let $w_i(x) = (M_1)_{i4}(x,k)$, then we can rewrite the fourth column of (2.88) as

$$w_i(x) = \delta_{i4} + \int_{-\infty}^{+\infty} \sum_{l=1}^{4} K(x,x',k)_{il} w_l(x',k) dx', \quad x \in \mathbb{R}, \quad k \in \bar{D}_1 \setminus \{0\}, \quad i = 1, \dots, 4, \tag{2.93}$$

where $K(x,x',k)$ is given by

$$K(x,x',k)_{il}=\begin{cases}H(x-x')\mathrm{e}^{(x-x')(l_i-l_4)}U_2(x',k)_{il}, & if \quad \gamma_{i4}^1=\gamma_1,\\ -H(x'-x)\mathrm{e}^{(x-x')(l_i-l_4)}U_2(x',k)_{il}, & if \quad \gamma_{i4}^1=\gamma_2.\end{cases} \tag{2.94}$$

Let $\varepsilon>0$ be small and define $\bar{D}_1^\varepsilon=\bar{D}_1\setminus\{|k|\le\varepsilon\}$. Then, there is a positive function $\tilde{b}_1(x)\in\mathcal{S}(\mathbb{R})$ such that

$$|U_2(x,k)|\le\tilde{b}_1(x),\quad x\in\mathbb{R},\quad k\in\bar{D}_1^\varepsilon.$$

Combining (2.94), we deduce that there exists $\tilde{b}(x)\in\mathcal{S}(\mathbb{R})$ such that the following holds:

$$\left|K(x,x',k)_{il}\right|\le\tilde{b}(x),\quad x,x'\in\mathbb{R},\quad k\in\bar{D}_1^\varepsilon,\quad i,l=1,2,3,4. \tag{2.95}$$

Set $K^{(0)}=1$ and define $K^{(m)}$ for $m\ge1$ by

$$K^{(m)}\begin{pmatrix}x_1,i_1,x_2,i_2,\dots,x_m,i_m\\ x_1',i_1',x_2',i_2',\dots,x_m',i_m'\end{pmatrix};\quad k\Bigg)=\det\begin{pmatrix}K\left(x_1,x_1',k\right)_{i_1i_1'} & \dots & K\left(x_1,x_m',k\right)_{i_1i_m'}\\ \vdots & & \vdots\\ K\left(x_m,x_1',k\right)_{i_mi_1'} & \dots & K\left(x_m,x_m',k\right)_{i_mi_m'}\end{pmatrix}. \tag{2.96}$$

Applying Hadamard's inequality, we can get

$$\left|K^{(m)}\begin{pmatrix}x_1,i_1,x_2,i_2,\dots,x_m,i_m\\ x_1',i_1',x_2',i_2',\dots,x_m',i_m'\end{pmatrix};\quad k\Bigg)\right|\le m^{\frac{m}{2}}\prod_{j=1}^m\tilde{b}\left(x_j'\right),\quad k\in\bar{D}_1^\varepsilon,\quad m\ge1. \tag{2.97}$$

Now let us define the Fredholm determinant $\tilde{f}(k)$ and Fredholm minor $\tilde{F}(x,x',k)$ related to (2.93):

$$\tilde{f}(k)=\sum_{m=0}^\infty\tilde{f}^{(m)}(k),\quad k\in\bar{D}_1^\varepsilon, \tag{2.98}$$

$$\tilde{F}(x,x',k)_{i,i'}=\sum_{m=0}^\infty\tilde{F}^{(m)}(x,x',k)_{ii'},\quad x,x'\in\mathbb{R},\quad k\in\bar{D}_1^\varepsilon,\quad i,i'=1,2,3,4, \tag{2.99}$$

where $\tilde{f}^{(m)}(k)$ and $\tilde{F}^{(m)}(x,x',k)_{ii'}$ are given by

$$\begin{aligned}\tilde{f}^{(m)}(k)&=\frac{(-1)^m}{m!}\sum_{i_1,i_2,\dots,i_m=1}^4\int_{\mathbb{R}^m}K^{(m)}\begin{pmatrix}x_1,i_1,x_2,i_2,\dots,x_m,i_m\\ x_1,i_1,x_2,i_2,\dots,x_m,i_m\end{pmatrix};\quad k\Bigg)dx_1dx_2\dots dx_m,\\ \tilde{F}^{(m)}(x,x',k)_{ii'}&=\frac{(-1)^m}{m!}\sum_{i_1,i_2,\dots,i_m=1}^4\\ &\quad\times\int_{\mathbb{R}^m}K^{(m+1)}\begin{pmatrix}x,i,x_1,i_1,x_2,i_2,\dots,x_m,i_m\\ x',i',x_1,i_1,x_2,i_2,\dots,x_m,i_m\end{pmatrix};\quad k\Bigg)dx_1dx_2\dots dx_m.\end{aligned} \tag{2.100}$$

It is easy to verify that $\tilde{f}^{(m)}(k)$ and $\tilde{F}^{(m)}(x,x',k)_{ii'}$ satisfy the estimates

$$\begin{aligned}\left|\tilde{f}^{(m)}(k)\right|&\le\frac{4^m m^{\frac{m}{2}}\left\|\tilde{b}(x)\right\|_{L^1(\mathbb{R})}^m}{m!},\quad k\in\bar{D}_1^\varepsilon,\quad m\ge0,\\ \left|\tilde{F}^{(m)}(x,x',k)_{ii'}\right|&\le\frac{4^m(m+1)^{\frac{(m+1)}{2}}\left\|\tilde{b}(x)\right\|_{L^1(\mathbb{R})}^m\tilde{b}(x')}{m!},\quad x,x'\in\mathbb{R}\quad k\in\bar{D}_1^\varepsilon,\quad m\ge0.\end{aligned} \tag{2.101}$$

Therefore, Fredholm determinant $\tilde{f}(k)$ and Fredholm minor $\tilde{F}(x,x',k)$ are well-defined for $x,x' \in \mathbb{R}$, $k \in \bar{D}_1^\varepsilon$. After completely similar analysis with Refs. [22] and [32], we derive that the solution of (2.93) can be written as

$$w_i(x) = \delta_{i4} + \frac{1}{\tilde{f}(k)} \int_{\mathbb{R}} \tilde{F}(x,x',k)_{i4} dx', \quad x \in \mathbb{R}, \quad k \in \bar{D}_1^\varepsilon \setminus \mathcal{Q}. \tag{2.102}$$

Then combined with estimations (2.101), we can easily infer that the assertions (a)–(d) are true for the fourth column of $M_1(x,k)$. Similar analysis can also be used for columns 1, 2, and 3, so we have proved the (a)–(d) for $n = 1$.

In order to prove (e), we need to give the specific form of γ_{ij}^1. For simplicity, we define the matrix γ, whose (i,j) element is exactly γ_{ij}^1. Then, it is easy to find

$$\gamma = \begin{pmatrix} \gamma_2 & \gamma_2 & \gamma_1 & \gamma_1 \\ \gamma_1 & \gamma_2 & \gamma_1 & \gamma_1 \\ \gamma_2 & \gamma_2 & \gamma_2 & \gamma_1 \\ \gamma_2 & \gamma_2 & \gamma_2 & \gamma_2 \end{pmatrix}, \tag{2.103}$$

where the definitions of γ_1 and γ_2 are shown in (2.89). So we have

$$M_1(x,k) \to \begin{pmatrix} 1 & 0 & * & * \\ * & 1 & * & * \\ 0 & 0 & 1 & * \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \text{as} \quad x \to +\infty. \tag{2.104}$$

On the other hand, we note that the trace of $U_2(x,k)$ is zero. According to the standard ODE theory, we conclude that $\det M_n(x,k)$ is independent of x. Then combined with (2.104), we get the assertion (e).

Symmetries (2.91) and (2.92) can be obtained according to the fact that the functions on both sides of these equations satisfy the same Fredholm integral equations, and then the left and right sides must be equal due to uniqueness. □

In the above proof, we only give the definition of the Fredholm determinant related to the fourth column of M_1. In fact, the Fredholm determinant $\tilde{f}_j(k)$ (note that $\tilde{f}(k) = \tilde{f}_4(k)$) associated with the jth $(1 \le j \le 4)$ column of M_1 can be defined as

$$\begin{aligned} \tilde{f}_j(k) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \sum_{i_1,i_2,\ldots,i_m=1}^{4} \\ \times \int_{\mathbb{R}^m} \det \begin{pmatrix} K_j(x_1,x_1,k)_{i_1 i_1} & \ldots & K_j(x_1,x_m,k)_{i_1 i_m} \\ \vdots & & \vdots \\ K_j(x_m,x_1,k)_{i_m i_1} & \ldots & K_j(x_m,x_m,k)_{i_m i_m} \end{pmatrix} dx_1 dx_2 \ldots dx_m, \end{aligned} \tag{2.105}$$

where

$$K_j(x,x',k)_{il} = \begin{cases} H(x-x') e^{(x-x')(l_i - l_j)} U_2(x',k)_{il}, & if \quad \gamma_{ij}^1 = \gamma_1, \\ -H(x'-x) e^{(x-x')(l_i - l_j)} U_2(x',k)_{il}, & if \quad \gamma_{ij}^1 = \gamma_2. \end{cases}$$

It is noted that each term of the above series satisfies the estimation similar to (2.101). Through the same analysis as $\tilde{f}(k)$, it is known that $\tilde{f}_j(k)$ is analytic in D_1 and can be continuously extended to $\bar{D}_1 \setminus \{0\}$. In addition, $\tilde{f}_j(k)$ does not vanish identically, so it has at most countable zeros in D_1. This fact implies that M_1 has at most countable poles in D_1.

Let us consider the asymptotic behavior of $M_n(x,k)$. As before, the asymptotic properties of $M_n(x,k)$ as $k \to \infty$ are obtained by considering the formal series:

$$M_{n,formal} = I + \frac{M_{n,1}(x)}{k} + \frac{M_{n,2}(x)}{k^2} + \cdots, \quad n = 1, \ldots, 8 \tag{2.106}$$

with boundary conditions

$$\begin{cases} \lim_{x\to-\infty}(M_{n,l}(x))_{i,j}=0, & if \quad \gamma_{ij}^n=\gamma_1, \\ \lim_{x\to+\infty}(M_{n,l}(x))_{i,j}=0, & if \quad \gamma_{ij}^n=\gamma_2. \end{cases} \tag{2.107}$$

Since $\gamma_{ii}^n=\gamma_2$ for $i=1,2,3,4$, $n=1,\dots,8$, we find that $M_{n,j}(x)=\mu_+^{[j]}(x)$ holds for all n and j. Following the same line of proof to [22, Lemma 4.3], the following lemma can be proved.

Lemma 2.8. *Suppose that $u_0,v_0\in\mathcal{S}(\mathbb{R})$ and $u_0,v_0\not\equiv 0$, then for any integer $j\geq 0$ and $p\geq 1$, there exists a constant $R\geq 0$ such that*

$$\left|\frac{\partial^j}{\partial k^j}\left(M(x,k)-I-\sum_{l=1}^{p}\frac{\mu_+^{[l]}(x)}{k^l}\right)\right|\leq\frac{C}{|k|^{p+1}},\quad x\in\mathbb{R},\quad k\in\mathbb{C}\setminus\Gamma,\quad |k|\geq R. \tag{2.108}$$

Remark 7. A similar proof can be found in [16, Theorem 8.4], although we define the function $M(x,k)$ in a different way.

Next, we will establish the relationship between $M(x,k)$ and $\mu_\pm(x,k)$. Through these relationships, we can obtain the asymptotic behavior of $M(x,k)$ as $k\to 0$. In order to describe our results, we introduce some notations.

Definition 2.3. For a given $n\times n$ matrix C, let

$$C\begin{pmatrix} i_1,i_2,\dots,i_p \\ k_1,k_2,\dots,k_p \end{pmatrix}=\det\begin{pmatrix} C_{i_1k_1} & C_{i_1k_2} & \dots & C_{i_1k_p} \\ C_{i_2k_1} & C_{i_2k_2} & \dots & C_{i_2k_p} \\ \vdots & \vdots & \ddots & \vdots \\ C_{i_pk_1} & C_{i_pk_2} & \cdots & C_{i_pk_p} \end{pmatrix}, \tag{2.109}$$

where $1\leq i_1<i_2<\dots<i_p\leq n$ and $1\leq k_1<k_2<\dots<k_p\leq n$. In particular, for principal minors of C, we further define

$$C\left(i_1,i_2,\dots,i_p\right)=C\begin{pmatrix} i_1,i_2,\dots,i_p \\ i_1,i_2,\dots,i_p \end{pmatrix}. \tag{2.110}$$

Lemma 2.9. *Suppose that $u_0,v_0\in\mathcal{S}(\mathbb{R})$ and have compact support. Then*

$$\begin{aligned} M_n(x,k)&=\mu_+(x,k)e^{x\hat{L}(k)}(T_n(k)) \\ &=\mu_-(x,k)e^{x\hat{L}(k)}(S_n(k)), \end{aligned} \tag{2.111}$$

where $S_n(k)$ and $T_n(k)$ are given by the elements of $s(k)$. We only give the expressions of $S_1(k)$, $S_8(k)$, $T_1(k)$, and $T_8(k)$, and the others can be obtained by similar calculations. The expressions of $S_1(k)$, $S_8(k)$, $T_1(k)$, and $T_8(k)$ are as follows:

$$S_1=\begin{pmatrix} \frac{s\left(1,2\right)}{s_{22}} & s_{12} & 0 & 0 \\ 0 & s_{22} & 0 & 0 \\ -\frac{s\begin{pmatrix}2,3\\1,2\end{pmatrix}}{s_{22}} & s_{32} & \frac{s\left(1,2,3\right)}{s\left(1,2\right)} & 0 \\ -\frac{s\begin{pmatrix}2,4\\1,2\end{pmatrix}}{s_{22}} & s_{42} & \frac{s\begin{pmatrix}1,2,4\\1,2,3\end{pmatrix}}{s\left(1,2\right)} & \frac{1}{s\left(1,2,3\right)} \end{pmatrix},\quad T_1=\begin{pmatrix} 1 & 0 & \frac{s\begin{pmatrix}1,2\\2,3\end{pmatrix}}{s\left(1,2\right)} & -\frac{s\begin{pmatrix}1,2,3\\2,3,4\end{pmatrix}}{s\left(1,2,3\right)} \\ -\frac{s_{12}}{s_{22}} & 1 & -\frac{s\begin{pmatrix}1,2\\1,3\end{pmatrix}}{s\left(1,2\right)} & \frac{s\begin{pmatrix}1,2,3\\1,3,4\end{pmatrix}}{s\left(1,2,3\right)} \\ 0 & 0 & 1 & -\frac{s\begin{pmatrix}1,2,3\\1,2,4\end{pmatrix}}{s\left(1,2,3\right)} \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{2.112}$$

$$S_8=\begin{pmatrix} \frac{s\begin{pmatrix}1,2,3\\2,3\end{pmatrix}}{s\begin{pmatrix}2,3\end{pmatrix}} & s_{12} & -\frac{s\begin{pmatrix}1,2\\2,3\end{pmatrix}}{s_{22}} & 0\\ 0 & s_{22} & 0 & 0\\ 0 & s_{32} & \frac{s\begin{pmatrix}2,3\end{pmatrix}}{s_{22}} & 0\\ \frac{s\begin{pmatrix}2,3,4\\1,2,3\end{pmatrix}}{s\begin{pmatrix}2,3\end{pmatrix}} & s_{42} & \frac{s\begin{pmatrix}2,4\\2,3\end{pmatrix}}{s_{22}} & \frac{1}{s\begin{pmatrix}1,2,3\end{pmatrix}} \end{pmatrix},\quad T_8=\begin{pmatrix} 1 & 0 & 0 & -\frac{s\begin{pmatrix}1,2,3\\2,3,4\end{pmatrix}}{s\begin{pmatrix}1,2,3\end{pmatrix}}\\ -\frac{s\begin{pmatrix}2,3\\1,3\end{pmatrix}}{s\begin{pmatrix}2,3\end{pmatrix}} & 1 & -\frac{s_{23}}{s_{22}} & \frac{s\begin{pmatrix}1,2,3\\1,3,4\end{pmatrix}}{s\begin{pmatrix}1,2,3\end{pmatrix}}\\ \frac{s\begin{pmatrix}2,3\\1,2\end{pmatrix}}{s\begin{pmatrix}2,3\end{pmatrix}} & 0 & 1 & -\frac{s\begin{pmatrix}1,2,3\\1,2,4\end{pmatrix}}{s\begin{pmatrix}1,2,3\end{pmatrix}}\\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{2.113}$$

Here, $s\left(i_1,\dots,i_p\right)$ *and* $s\begin{pmatrix}i_1,\dots,i_p\\k_1,\dots,k_p\end{pmatrix}$ *follow Definition 2.3.*

Proof. Note that u_0 and v_0 are compactly supported in $\mathbb{R}$, then $\mu_\pm(x,k)$ and $s(k)$ are analytic in $\mathbb{C}\setminus\{0\}$. Since both $\mu_\pm(x,k)$ and $M_n(x,k)$ are fundamental solutions of (2.13), there exist matrices $S_n(k)$ and $T_n(k)$ independent of x such that

$$M_n(x,k)=\mu_+(x,k)\mathrm{e}^{x\hat{L}(k)}(T_n(k))=\mu_-(x,k)\mathrm{e}^{x\hat{L}(k)}(S_n(k)),\quad k\in\bar{D}_n\setminus\mathcal{Q}. \tag{2.114}$$

Furthermore, we find that when x is sufficiently large, $\mu_+(x,k)$ are equal to $I_{4\times4}$. So we have

$$\begin{cases} S_n(k)=\lim_{x\to-\infty}e^{-x\hat{L}(k)}(M_n(x,k)),\\ T_n(k)=\lim_{x\to+\infty}e^{-x\hat{L}(k)}(M_n(x,k)), \end{cases}\quad k\in\bar{D}_n\setminus\mathcal{Q}. \tag{2.115}$$

Indeed, from (2.88) we deduce that when u_0 and v_0 have compact support and $|x|$ is sufficiently large, $\mathrm{e}^{-x\hat{L}(k)}(M_n(x,k))$ does not depend on x, so the two limits in (2.115) exist.

Now let us derive the exact forms of $S_1(k)$ and $T_1(k)$. Combining the assertion (g) of Proposition 2.4 and Equation (2.114), we obtain

$$s(k)=S_1(k)T_1^{-1}(k),\quad k\in\bar{D}_n\setminus\mathcal{Q}. \tag{2.116}$$

On the other hand, by (2.115) and integral equation (2.88) we deduce that

$$(S_1(k))_{ij}=0,\quad if\quad \gamma_{ij}^1=\gamma_1,$$

$$(T_1(k))_{ij}=\delta_{ij},\quad if\quad \gamma_{ij}^1=\gamma_2.$$

Therefore, the relation (2.116) is actually a system of 16 linear equations with 16 unknowns, and the expressions of $S_1(k)$ and $T_1(k)$ are obtained by solving the algebraic system. Of course, we can obtain $S_n(k)$ and $T_n(k)$, $n=2,\dots,8$ by similar calculations. □

As in Section 2.5, let $u_0^{(i)}$ and $v_0^{(i)}$ be defined by (2.69). Then we give a result similar to Lemma 2.5.

Lemma 2.10. *Assume* $u_0,v_0\in\mathcal{S}(\mathrm{R})$. *Suppose the scattering matrices and eigenfunctions associated with the initial data* (u_0,v_0) *and* $(u_0^{(i)},v_0^{(i)})$ *are denoted by* $\left\{s(k),s^A(k),M_n(x,k)\right\}$ *and* $\left\{s^{(i)}(k),(s^A)^{(i)}(k),M_n^{(i)}(x,k)\right\}$, *respectively. Then*

$$\lim_{i\to\infty} s^{(i)}(k) = s(k), \qquad k \in \begin{pmatrix} -\mathrm{i}\bar{S} & \mathbb{R}_+ e^{\frac{\pi}{4}\mathrm{i}} & \emptyset & \mathbb{R}_+ e^{\frac{3\pi}{4}\mathrm{i}} \\ \mathbb{R}_+ e^{\frac{\pi}{4}\mathrm{i}} & -\bar{S} & \mathbb{R}_+ e^{-\frac{\pi}{4}\mathrm{i}} & \emptyset \\ \emptyset & \mathbb{R}_+ e^{-\frac{\pi}{4}\mathrm{i}} & \mathrm{i}\bar{S} & -\mathbb{R}_+ e^{\frac{\pi}{4}\mathrm{i}} \\ \mathbb{R}_+ e^{\frac{3\pi}{4}\mathrm{i}} & \emptyset & -\mathbb{R}_+ e^{\frac{\pi}{4}\mathrm{i}} & \bar{S} \end{pmatrix} \setminus \{0\}, \tag{2.117}$$

$$\lim_{i\to\infty} (s^A)^{(i)}(k) = s^A(k), \qquad k \in \begin{pmatrix} \mathrm{i}\bar{S} & \mathbb{R}_+ e^{\frac{-3\pi}{4}\mathrm{i}} & \emptyset & \mathbb{R}_+ e^{-\frac{\pi}{4}\mathrm{i}} \\ \mathbb{R}_+ e^{-\frac{3\pi}{4}\mathrm{i}} & \bar{S} & \mathbb{R}_+ e^{\frac{3\pi}{4}\mathrm{i}} & \emptyset \\ \emptyset & \mathbb{R}_+ e^{\frac{3\pi}{4}\mathrm{i}} & -\mathrm{i}\bar{S} & \mathbb{R}_+ e^{\frac{\pi}{4}\mathrm{i}} \\ \mathbb{R}_+ e^{-\frac{\pi}{4}\mathrm{i}} & \emptyset & \mathbb{R}_+ e^{\frac{\pi}{4}\mathrm{i}} & -\bar{S} \end{pmatrix} \setminus \{0\}, \tag{2.118}$$

$$\lim_{i\to\infty} M_n^{(i)}(x,k) = M_n(x,k), \qquad x \in \mathbb{R},\ k \in \bar{D}_n \setminus \mathcal{Q}, \quad n = 1,2,\dots,7,8. \tag{2.119}$$

Proof. The proof for this lemma is similar to the proof of [22, Lemma 4.5]. We omit the details for brevity. □

In the case that u_0 and v_0 are compactly supported, Lemma 2.9 establishes the relationship between $M(x,k)$ and $\mu_\pm(x,k)$. Our following lemma gives the results in the general case.

Lemma 2.11. *Let $u_0, v_0 \in \mathcal{S}(\mathbb{R})$ and define $M_n^A(x,k) = (M_n^{-1})^\top(x,k)$. Then we have*

$$\begin{cases} M_1 = \begin{pmatrix} -\dfrac{L(\mu_{-2}^A, \mu_{+3}^A, \mu_{+4}^A)}{s_{22}} & \mu_{+2} & -\dfrac{L(\mu_{-1}^A, \mu_{-2}^A, \mu_{+4}^A)}{\Delta} & \dfrac{\mu_{-4}}{s_{44}^A} \end{pmatrix}, \\ M_1^A = \begin{pmatrix} -\dfrac{L(\mu_{+2}, \mu_{-3}, \mu_{-4})}{\Delta} & \dfrac{\mu_{-2}^A}{s_{22}} & -\dfrac{L(\mu_{+1}, \mu_{+2}, \mu_{-4})}{s_{44}^A} & \mu_{+4}^A \end{pmatrix}, \end{cases} \tag{2.120}$$

$$\begin{cases} M_8 = \begin{pmatrix} -\dfrac{L(\mu_{-2}^A, \mu_{-3}^A, \mu_{+4}^A)}{\Xi} & \mu_{+2} & -\dfrac{L(\mu_{+1}^A, \mu_{-2}^A, \mu_{+4}^A)}{s_{22}} & \dfrac{\mu_{-4}}{s_{44}^A} \end{pmatrix}, \\ M_8^A = \begin{pmatrix} -\dfrac{L(\mu_{+2}, \mu_{+3}, \mu_{-4})}{s_{44}^A} & \dfrac{\mu_{-2}^A}{s_{22}} & -\dfrac{L(\mu_{-1}, \mu_{+2}, \mu_{-4})}{\Xi} & \mu_{+4}^A \end{pmatrix}, \end{cases} \tag{2.121}$$

where Δ and Ξ are given by (2.64) and (2.65).

Proof. Our strategy is to first prove that (2.120) and (2.121) hold in the case of compactly supported potentials $(u_0^{(i)}, v_0^{(i)})$, where $(u_0^{(i)}, v_0^{(i)})$ are determined by (u_0, v_0) according to (2.69). Then take $i \to \infty$, and apply Lemma 2.5, Corollary 2.2, and Lemma 2.10 to get the desired result. In the following proof, we assume that (u_0, v_0) have compact support. Based on the above analysis, it will suffice to prove this special case.

In this proof, we denote the jth column of a given matrix A as $[A]_j$. Then according to (2.111), we get

$$[M_1]_2 = \mu_{+2}, \qquad [M_1]_4 = \frac{\mu_{-4}}{s_{44}^A}.$$

On the other hand, note that

$$M_1^A = \mu_+^A e^{-x\hat{L}(k)} \left(T_1(k)^{-1}\right)^\top = \mu_-^A e^{-x\hat{L}(k)} \left(S_1(k)^{-1}\right)^\top, \tag{2.122}$$

then by direct calculation we obtain

$$\left[\left(M_1^A\right)\right]_2 = \frac{\mu_{-2}^A}{s_{22}}, \qquad \left[\left(M_1^A\right)\right]_4 = \mu_{+4}^A.$$

Therefore, we may write M_1 and M_1^A as

$$M_1 = \begin{pmatrix} \chi_1 & \mu_{+2} & \chi_3 & \dfrac{\mu_{-4}}{s_{44}^A} \end{pmatrix}, \qquad M_1^A = \begin{pmatrix} \tilde{\chi}_1 & \dfrac{\mu_{-2}^A}{s_{22}} & \tilde{\chi}_3 & \mu_{+4}^A \end{pmatrix}, \tag{2.123}$$

where χ_1, χ_3, $\tilde{\chi}_1$, and $\tilde{\chi}_3$ are to be determined. Then, apply Lemma 2.7 we have

$$\chi_1 = -L\left(\frac{\mu_{-2}^A}{s_{22}}, \tilde{\chi}_3, \mu_{+4}^A\right), \tag{2.124}$$

$$\tilde{\chi}_1 = -L\left(\mu_{+2}, \chi_3, \frac{\mu_{-4}}{s_{44}^A}\right). \tag{2.125}$$

Define $\Delta(k) = \det \begin{pmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{pmatrix}$, again by Equation (2.111) we get

$$\chi_3 = \frac{s_{44}^A}{\Delta}\mu_{-3} - \mathrm{e}^{x(l_4-l_3)}\frac{s_{34}^A}{\Delta}\mu_{-4}, \tag{2.126}$$

$$\chi_1 = \mu_{+1} - \mathrm{e}^{x(l_2-l_1)}\frac{s_{21}}{s_{22}}\mu_{+2}. \tag{2.127}$$

Substituting the expression (2.126) of χ_3 into the expression (2.125) for $\tilde{\chi}_1$, we obtain

$$\tilde{\chi}_1 = -\frac{1}{\Delta}L(\mu_{+2}, \mu_{-3}.\mu_{-4}). \tag{2.128}$$

Next let us find a suitable expression for χ_1. Again by Lemma 2.7, we know that

$$\tilde{\chi}_3 = -L\left(\chi_1, \mu_{+2}, \frac{\mu_{-4}}{s_{44}^A}\right). \tag{2.129}$$

Then substituting the expression (2.127) of χ_1 into the expression (2.129), we arrive at

$$\tilde{\chi}_3 = -\frac{1}{s_{44}^A}L(\mu_{+1}, \mu_{+2}, \mu_{-4}). \tag{2.130}$$

On the other hand, from the assertion (g) in Proposition 2.4 and the equation $s^A = (s^{-1})^\top$, it is easy to verify

$$\mu_{-4} = \mathrm{e}^{x(l_1-l_4)}s_{41}^A\mu_{+1} + \mathrm{e}^{x(l_2-l_4)}s_{42}^A\mu_{+2} + \mathrm{e}^{x(l_3-l_4)}s_{43}^A\mu_{+3} + s_{44}^A\mu_{+4}. \tag{2.131}$$

Then the above expression is brought into Equation (2.130) to obtain

$$\begin{aligned}\tilde{\chi}_3 =& -L(\mu_{+1}, \mu_{+2}, \mu_{+4}) - \mathrm{e}^{x(l_3-l_4)}\frac{s_{43}^A}{s_{44}^A}L(\mu_{+1}, \mu_{+2}, \mu_{+3}) \\ =& \mu_{+3}^A - \mathrm{e}^{x(l_3-l_4)}\frac{s_{43}^A}{s_{44}^A}\mu_{+4}^A.\end{aligned} \tag{2.132}$$

Finally, combining Equations (2.124) and (2.132), we conclude that

$$\chi_1 = -\frac{1}{s_{22}}L\left(\mu_{-2}^A, \mu_{+3}^A, \mu_{+4}^A\right). \tag{2.133}$$

Now we have obtained the exact expressions of M_1 and M_1^A except for the third column χ_3 of M_1. It is not difficult to obtain the expression of χ_3 following the way we get χ_1. We omit the details. Similarly, the expressions of M_8 and M_8^A can be obtained by the same technique. □

Lemma 2.11 shows that, except for the origin, the other singularities of $M_1(x,k)$ come from the zeros of $s_{22}(k)$, $s^A_{44}(k)$, and $\Delta(k)$. Thus we figured out the members of the set $\mathcal{Q}\cap D_1$. Furthermore, by similar calculations we are able to obtain that $M_2,\dots,M_7$ have similar expressions to (2.120), and thus we are able to determine the set $\mathcal{Q}$.

3 | The RH Problem

In the previous section, we have constructed the solution $M(x,k)$ of the fundamental meromorphic function of Equation (2.13) from initial data u_0, $v_0\in\mathcal{S}(\mathbb{R})$. In this section, we consider the effect of time evolution and define a fundamental meromorphic function solution $M(x,t,k)$ of Lax pair (2.10). We derive the RH problem that $M(x,t,k)$ satisfies, and we will show how to reconstruct the potentials $\{u,v\}$ according to the solution of the RH problem.

We first give the jump condition that $M(x,k)$ satisfies. Introduce the matrix $\Theta(x,t,k)$ by

$$\Theta(x,t,k)=xL(k)+tZ(k).$$

Let the reflection coefficients $\rho(k)$, $r_1(k)$, and $r_2(k)$ be defined by

$$\rho(k)=\frac{\Lambda(k)}{\Delta(k)},\qquad k\in\mathbb{R}_+;\qquad\qquad \begin{cases} r_1(k)=\dfrac{s_{12}(k)}{s_{11}(k)},\\ r_2(k)=\dfrac{s^A_{34}(k)}{s^A_{33}(k)}, \end{cases}\qquad k\in\mathbb{R}_+e^{\frac{\pi}{4}\mathrm{i}}. \tag{3.1}$$

It easy to verify that the reflection coefficients r_1 and r_2 satisfy $r_2(k)=-\mathrm{i}r_1(k)$ by using the symmetries of the scattering matrix. Moreover, for $k\in\Gamma$ we define the matrix $v(k)$ as follows:

$$v_1=\begin{pmatrix} 1 & 0 & \rho(k) & 0\\ 0 & 1 & 0 & 0\\ -\rho^*(k) & 0 & 1-|\rho(k)|^2 & 0\\ 0 & 0 & 0 & 1 \end{pmatrix},\quad v_2=\begin{pmatrix} 1 & -r_1(k) & 0 & 0\\ r_1^*(k) & 1-|r_1(k)|^2 & 0 & 0\\ 0 & 0 & 1-|r_2(k)|^2 & -r_2^*(k)\\ 0 & 0 & r_2(k) & 1 \end{pmatrix},$$

$$v_3=\begin{pmatrix} 1 & 0 & 0 & 0\\ 0 & 1-|\rho(-\mathrm{i}k)|^2 & 0 & -\rho^*(-\mathrm{i}k)\\ 0 & 0 & 1 & 0\\ 0 & \rho(-\mathrm{i}k) & 0 & 1 \end{pmatrix},\quad v_4=\begin{pmatrix} 1-|r_1(-\mathrm{i}k)|^2 & 0 & 0 & r_1^*(-\mathrm{i}k)\\ 0 & 1-|r_2(-\mathrm{i}k)|^2 & -r_2^*(-\mathrm{i}k) & 0\\ 0 & r_2(-\mathrm{i}k) & 1 & 0\\ -r_1(-\mathrm{i}k) & 0 & 0 & 1 \end{pmatrix},$$

$$v_5=\begin{pmatrix} 1-|\rho(-k)|^2 & 0 & -\rho^*(-k) & 0\\ 0 & 1 & 0 & 0\\ \rho(-k) & 0 & 1 & 0\\ 0 & 0 & 0 & 1 \end{pmatrix},\quad v_6=\begin{pmatrix} 1-|r_2(-k)|^2 & -r_2^*(-k) & 0 & 0\\ r_2(-k) & 1 & 0 & 0\\ 0 & 0 & 1 & -r_1(-k)\\ 0 & 0 & r_1^*(-k) & 1-|r_1(-k)|^2 \end{pmatrix},$$

$$v_7=\begin{pmatrix} 1 & 0 & 0 & 0\\ 0 & 1 & 0 & \rho(\mathrm{i}k)\\ 0 & 0 & 1 & 0\\ 0 & -\rho^*(\mathrm{i}k) & 0 & 1-|\rho(\mathrm{i}k)|^2 \end{pmatrix},\quad v_8=\begin{pmatrix} 1 & 0 & 0 & r_2(\mathrm{i}k)\\ 0 & 1 & -r_1(\mathrm{i}k) & 0\\ 0 & r_1^*(\mathrm{i}k) & 1-|r_1(\mathrm{i}k)|^2 & 0\\ -r_2^*(\mathrm{i}k) & 0 & 0 & 1-|r_2(\mathrm{i}k)|^2 \end{pmatrix},$$

where $v_j(k)$ denotes the restriction of $v(k)$ to Γ_j.

Then we have the following lemma.

Lemma 3.1. *Assume that $u_0, v_0 \in \mathcal{S}(\mathbb{R})$. Then for each $x \in \mathbb{R}$, when k approaches Γ from the left or right, the limits of $M(x,k)$ exist, denoted by $M_+(x,k)$ and $M_-(x,k)$. Furthermore, $M_+(x,k)$ and $M_-(x,k)$ satisfy the jump conditions:*

$$M_+(x,k) = M_-(x,k)e^{xL(k)}v(k)e^{-xL(k)}, \qquad k \in \Gamma \setminus \mathcal{Q}. \tag{3.2}$$

Proof. We will show that

$$M_1 = M_8 e^{xL(k)} v_1 \mathrm{e}^{-xL(k)}, \qquad k \in \mathbb{R}_+ \setminus \mathcal{Q}. \tag{3.3}$$

The proof of the remaining part is similar. As in the proof to Lemma 2.11, we only need to prove that the above jump relation holds for the compact supported potentials. Since both $M_1\mathrm{e}^{xL(k)}$ and $M_8\mathrm{e}^{xL(k)}$ are fundamental solutions of differential equation $\phi_x = F\phi$, then there exists a matrix K independent of x such that

$$M_1(x,k) = M_8(x,k)\mathrm{e}^{xL(k)}K(k)\mathrm{e}^{-xL(k)}, \qquad k \in \mathbb{R}_+ \setminus \mathcal{Q}. \tag{3.4}$$

Since u_0 and v_0 are compactly supported, by Lemma 2.9 we have

$$M_n(x,k) = \mathrm{e}^{xL(k)}S_n(k)\mathrm{e}^{-xL(k)}, \qquad x < -b_0, \tag{3.5}$$

where b_0 is a sufficiently large positive constant. Thus, we get

$$K(k) = S_8(k)^{-1}S_1(k).$$

A straightforward calculation yields (3.3). □

Next, we consider the evolution of scattering data over time. Assume that $\{u(x,t), v(x,t)\}$ is a Schwartz class solution of system (1.1) with existence time $T \in (0, +\infty]$ and initial data $\{u_0, v_0\} \in \mathcal{S}(\mathbb{R})$. We replace $\{u_0, v_0\}$ with $\{u(x,t), v(x,t)\}$ in all equations containing $u_0(x)$ and $v_0(x)$ in the previous sections. For example, we can define $\mu_\pm(x,t,k)$, $\mu_\pm^A(x,t,k)$, $s(t,k)$, $s^A(t,k)$, $f_n(x,t,k)$, $g_n(x,t,k)$, and $M(x,t,k)$ by doing this in Equations (2.13), (2.39), (2.33), (2.46), (2.56), (2.57), and (2.88). From our definition of $M(x,t,k)$, it is easy to see that $M(x,k) = M(x,0,k)$. Since $M(x,k)$ satisfies Equation (2.13), one is not surprised that $M(x,t,k)$ satisfies Lax pair (2.10). We give the following lemma, the proof of which can be found in [22, Lemma 5.1].

Lemma 3.2. *Let $n \in \{1, \dots, 8\}$. For $k \in \bar{D}_n \setminus \{0\}$ and $(x,t) \in (\mathbb{R}, (0,T])$, $M_n(x,t,k)$ satisfies the Lax pair equations (2.10).*

As a consequence of this lemma, we can derive the jump conditions that $M(x,t,k)$ satisfies.

Lemma 3.3. *Assume that $u_0, v_0 \in \mathcal{S}(\mathbb{R})$. Then for each $x \in \mathbb{R}$, when k approaches Γ from the left or right, the limits of $M(x,t,k)$ exist, denoted by $M_+(x,t,k)$ and $M_-(x,t,k)$. Furthermore, $M_+(x,t,k)$ and $M_-(x,t,k)$ satisfy the jump condition:*

$$M_+(x,t,k) = M_-(x,t,k)e^{\Theta(x,t,k)}v(k)e^{-\Theta(x,t,k)}, \qquad k \in \Gamma \setminus \mathcal{Q}. \tag{3.6}$$

Proof. We only show that

$$M_1 = M_8\mathrm{e}^{\Theta(x,t,k)}v_1(k)\mathrm{e}^{-\Theta(x,t,k)}, \qquad k \in \mathbb{R}_+ \setminus \mathcal{Q}.$$

Since M_1 and M_8 satisfy (2.10), then $M_1\mathrm{e}^{\Theta(x,t,k)}$ and $M_8\mathrm{e}^{\Theta(x,t,k)}$ satisfy (2.4). These equations imply that the functions M_1 and M_8 are related by

$$M_1(x,t,k) = M_8(x,t,k)\mathrm{e}^{\Theta(x,t,k)}(M_8(0,0,k)^{-1}M_1(0,0,k))\mathrm{e}^{-\Theta(x,t,k)}. \tag{3.7}$$

From Lemma 3.1, we know that $M_8(0,0,k)^{-1}M_1(0,0,k) = v_1(k)$. □

Let us next show the evolution of the scattering matrix over time.

Lemma 3.4. *Assume that $\{u(x,t), v(x,t)\}$ is a Schwart class solution of system (1.1) with the initial data $\{u_0, v_0\}$. Let $s(k,t)$ be the scattering matrix associated with $\{u(x,t), v(x,t)\}$, and let $s(k)$ be the scattering matrix associated with $\{u_0, v_0\}$. Then*

we have

$$s(k,t) = e^{tZ(k)}s(k)e^{-tZ(k)}. \tag{3.8}$$

Proof. As before, it is sufficient to prove the lemma for compactly supported potentials. Under the assumption of this lemma, as a standard result of inverse scattering analysis, one can verify that $\mu_\pm(x,t,k)\mathrm{e}^{\Theta(x,t,k)}$ are simultaneous matrix solutions of the compatible linear problems (2.4). Thus we have

$$\mu_+(x,t,k)\mathrm{e}^{\Theta(x,t,k)} = \mu_-(x,t,k)\mathrm{e}^{\Theta(x,t,k)}(\mu_-(0,0,k)^{-1}\mu_+(0,0,k)) = \mu_-(x,t,k)\mathrm{e}^{\Theta(x,t,k)}s(k). \tag{3.9}$$

On the other hand, recall that $s(k,t)$ and $\mu_\pm(x,t,k)$ are defined by the following integral equations:

$$s(k,t) = I - \int_{\mathbb{R}} \mathrm{e}^{-x\hat{L}(k)}(U_2\mu_+)(x,t,k)dx,$$

$$\mu_\pm(x,t,k) = I + \int_{\pm\infty}^{x} \mathrm{e}^{(x-y)\hat{L}(k)}(U_2\mu_\pm)(y,t,k)dy.$$

Noting that $\{u(x,t), v(x,t)\}$ have compact support, we obtain

$$\mu_+(x,t,k)\mathrm{e}^{xL(k)} = \mu_-(x,t,k)\mathrm{e}^{xL(k)}s(k,t).$$

Now combining the above equation with (3.9), we immediately get (3.8). □

Remark 8. From this lemma, we can obtain the time evolution of the reflection coefficients. Let $\{\rho(k,t), r_1(k,t), r_2(k,t)\}$ be the reflection coefficients associated to $\{u(x,t), v(x,t)\}$. Then we have

$$\rho(k,t) = \rho(k)\mathrm{e}^{\theta_{13}(0,t,k)}, \qquad r_1(k,t) = r_1(k)\mathrm{e}^{\theta_{12}(0,t,k)}, \qquad r_2(k,t) = r_2(k)\mathrm{e}^{\theta_{43}(0,t,k)}, \tag{3.10}$$

where $\theta_{ij}(x,t,k)$ is defined as $\theta_{ij}(x,t,k) = x(l_i - l_j) + t(z_i - z_j)$.

Remark 9. By similar calculations, we can obtain the time evolution of the adjoint scattering matrix, that is, $s^A(k,t) = \mathrm{e}^{-tZ(k)}s^A(k)\mathrm{e}^{tZ(k)}$.

In order to obtain the RH problem satisfied by $M(x,t,k)$, we have to analyze the asymptotic behavior of $M(x,t,k)$ as $k \to \infty$ and $k \to 0$. There is no doubt that when $k \to \infty$ the asymptotic expansion (2.108) still holds if we replace $\{\mu_+^{[l]}(x)\}_{l=1}^{\infty}$ by $\{\mu_+^{[l]}(x,t)\}_{l=1}^{\infty}$, where $\{\mu_+^{[l]}(x,t)\}_{l=1}^{\infty}$ are the asymptotic coefficients of $\mu_+(x,t,k)$ as $k \to \infty$. But when $k \to 0$ the situation is more complicated. We did not even give the corresponding asymptotic description at $t = 0$ in the previous sections. But as we mentioned earlier, Lemma 2.11 establishes the connection of $M(x,k)$ with $\mu_\pm(x,k)$, whose asymptotic behavior as $k \to 0$ is known. Therefore, we can determine the asymptotic behavior of $M(x,k)$ according to the asymptotic expansions of $\mu_\pm(x,k)$. In the following, we consider two special cases that interest us: (1) Soliton-free case; and (2) Pure soliton case. In both cases, we will add some assumptions to ensure that the calculations are more convenient.

3.1 | Soliton-Free Case

In this subsection, we consider the case that the discrete scattering data do not contribute to the potentials, that is, the set $\mathcal{Z}$ defined in Section 2.6 is an empty set. Therefore, we require the following assumption to be true.

Assumption 3.1. Assume that $s_{22}(k)$, $s_{44}^A(k)$, $\Delta(k)$ have no zeros for $k \in \bar{D}_1 \setminus \{0\}$.

Moreover, we will impose additional assumption on the asymptotic behavior of these spectral functions as $k \to 0$. For simplicity, we will be limited to the general case, that is, the following assumption is satisfied.

Assumption 3.2. Assume that

$$\lim_{k\to 0} ks_{22}(k) \neq 0, \qquad \lim_{k\to 0} ks_{44}^A(k) \neq 0, \qquad \lim_{k\to 0} k^2\Delta(k) \neq 0.$$

Remark 10. Due to symmetries (2.49) and (2.35), one can easily verify that $s_{22}(k) = s_{44}^A(k)$. Thus, it suffices to require $s_{22}(k)$ to satisfy Assumptions 3.1 and 3.2.

Now we are ready to describe the asymptotic behavior of $M(x,t,k)$ when k tends to 0.

Lemma 3.5. *Assume that $u(x,t)$ and $v(x,t)$ are the Schwart class solutions of system (1.1), and the initial data $\{u_0, v_0\}$ make Assumption 3.2 hold. Then for any integer $p \geq 1$, there exist matrices $\left\{\mathcal{M}_n^{(l)}(x,t)\right\}_{l=-1}^{p}$ satisfying the following estimates:*

$$\left| M_n(x,t,k) - \sum_{l=-1}^{p} \mathcal{M}_n^{(l)}(x,t)k^l \right| \leq C|k|^{p+1}, \quad |k| \leq 2, \quad k \in \bar{D}_n \setminus \{0\}, \quad n = 1,\dots,8. \tag{3.11}$$

In particular, $\mathcal{M}_1^{(-1)}(x,t)$ has the form:

$$\mathcal{M}_1^{(-1)}(x,t) = \begin{pmatrix} -\mathrm{i}\delta_1 & -\mathrm{i}\alpha_+ & 0 & 0 \\ \delta_2 & -\beta_+ & 0 & 0 \\ \mathrm{i}\delta_1 & \mathrm{i}\alpha_+ & 0 & 0 \\ -\delta_2 & \beta_+ & 0 & 0 \end{pmatrix},$$

where δ_1, δ_2, α_+, and β_+ are some scalar functions of x and t.

Proof. We only need to calculate the asymptotic behavior of M_1 when $k \to 0$. Then according to symmetry (2.91), the asymptotic properties of M_n can be determined for each $n \in \{2,3,\dots,8\}$. From (2.120), we know that $M_1(x,k)$ can be expressed by $\mu_\pm(x,k)$, $\mu_\pm^A(x,k)$, $s(k)$, and $s^A(k)$. Lemma 3.4, Remark 9, and Equation (2.82) show that

$$s_{22}(k,t) = s_{22}(k), \qquad s_{44}^A(k,t) = s_{44}^A(k), \qquad \Delta(k,t) = \Delta(k).$$

Then along the lines of the proof of Lemma 2.11, it is not difficult to obtain

$$M_1(x,t,k) = \left(-\frac{L(\mu_{-2}^A(x,t,k), \mu_{+3}^A(x,t,k), \mu_{+4}^A(x,t,k))}{s_{22}(k)} \quad \mu_{+2}(x,t,k) \quad -\frac{L(\mu_{-1}^A(x,t,k), \mu_{-2}^A(x,t,k), \mu_{+4}^A(x,t,k))}{\Delta(k)} \quad \frac{\mu_{-4}(x,t,k)}{s_{44}^A(k)} \right). \tag{3.12}$$

On the other hand, the asymptotic behavior of $\mu_\pm(x,k)$, $\mu_\pm^A(x,k)$, $s(k)$, and $s^A(k)$ as $k \to 0$ has been calculated in Propositions 2.3, 2.6, 2.4, and 2.8. According to their definitions, one can easily verify that the asymptotic expansions of $\mu_\pm(x,t,k)$ and $\mu_\pm^A(x,t,k)$ have the same form as those of $\mu_\pm(x,k)$ and $\mu_\pm^A(x,k)$. Bringing these expansions into (3.12) and combining with Assumption 3.2, we complete the proof. □

We now study the properties of the reflection coefficients.

Theorem 3.1. *Suppose that $u_0, v_0 \in S(\mathbb{R})$ satisfy Assumptions 3.1 and 3.2. Define $\mathbb{R}_+ = (0, +\infty)$. Then the reflection coefficients $\rho(k)$, $r_1(k)$, and $r_2(k)$ are well-defined by (3.1) and have the following properties:*

a. $r_1(k), r_2(k) \in C^\infty(\mathbb{R}_+ \mathrm{e}^{\frac{\pi}{4}\mathrm{i}})$ and $\rho(k) \in C^\infty(\mathbb{R}_+)$.

b. $r_1(k)$, $r_2(k)$, and $\rho(k)$ are rapidly decreasing as $|k| \to \infty$.

c. The functions $\rho(k)$, $r_1(k)$, and $r_2(k)$ have continuous boundary values at $k = 0$.

d. $|r_1(k)| < 1$ and $|r_2(k)| < 1$ hold for all $k \in \mathbb{R}_+ \mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$, and $|\rho(k)| < 1$ for all $k \in \mathbb{R}_+$.

Proof. The statement (a) follows from Propositions 2.4, 2.8, 2.9 and Lemma 2.4. From Lemma 2.8, we know that $\frac{\partial^j}{\partial k^j}M_+(0,k)$ and $\frac{\partial^j}{\partial k^j}M_-(0,k)$ have the same asymptotic expansion as $k\to\infty$. Combining this with (3.2) yields statement (b).

To prove (c) and (d), we first review some notations used in Section 2. Assume that $\{u(x,t),v(x,t)\}$ are the Schwart class solutions of system (1.1). Let the compactly supported potentials $\{u^{(i)}(x,t),v^{(i)}(x,t)\}$ be defined by (2.69). And let $s(k)$ and $s^{(i)}(k)$ denote the scattering matrices associated with $\{u,v\}$ and $\{u^{(i)},v^{(i)}\}$, respectively. The reflection coefficients associated with $\{u^{(i)}(x,0),v^{(i)}(x,0)\}$ are denoted as $\rho^{(i)}(k)$, $r_1^{(i)}(k)$, and $r_2^{(i)}(k)$. By Corollary 2.3 and Lemma 2.10, we have

$$\begin{aligned}
&\Delta(k)=\lim_{i\to\infty}\Delta^{(i)}(k),\qquad \Lambda(k)=\lim_{i\to\infty}\Lambda^{(i)}(k),\qquad k\in\mathbb{R}_+;\\
&s_{12}(k)=\lim_{i\to\infty}s_{12}^{(i)}(k),\qquad s_{11}(k)=\lim_{i\to\infty}s_{11}^{(i)}(k),\qquad \mathbb{R}_+e^{\frac{\pi}{4}i};\\
&s_{34}^{A}(k)=\lim_{i\to\infty}s_{34}^{A(i)}(k),\qquad s_{33}^{A}(k)=\lim_{i\to\infty}s_{33}^{A(i)}(k),\qquad k\in\mathbb{R}_+e^{\frac{\pi}{4}i};
\end{aligned}\tag{3.13}$$

where

$$\Delta^{(i)}(k)=\det\begin{pmatrix}s_{11}^{(i)}&s_{12}^{(i)}\\ s_{21}^{(i)}&s_{22}^{(i)}\end{pmatrix},\qquad \Lambda^{(i)}(k)=\lim_{i\to\infty}\det\begin{pmatrix}s_{12}^{(i)}&s_{13}^{(i)}\\ s_{22}^{(i)}&s_{23}^{(i)}\end{pmatrix}.$$

Based on these relations, we conclude that $\{u^{(i)}(x,t),v^{(i)}(x,t)\}$ must satisfy Assumptions 3.1 and 3.2 when i is sufficiently large. Therefore, we may assume that $\{u^{(i)}(x,t),v^{(i)}(x,t)\}$ satisfy Assumptions 3.1 and 3.2 for each i. Moreover, by symmetries (2.35), $s^{(i)}(k)$ has an asymptotic expansion of the following form as $k\to 0$:

$$s^{(i)}(k)=\frac{s_{-1}^{(i)}}{k}+O(1),\qquad s_{-1}^{(i)}=\begin{pmatrix}\mathfrak{s}_1^{(i)}&\mathfrak{s}_2^{(i)}&\mathfrak{s}_3^{(i)}&-\mathfrak{s}_2^{*(i)}\\ i\mathfrak{s}_2^{*(i)}&-i\mathfrak{s}_1^{(i)}&-i\mathfrak{s}_2^{(i)}&-i\mathfrak{s}_3^{(i)}\\ -\mathfrak{s}_3^{(i)}&\mathfrak{s}_2^{*(i)}&-\mathfrak{s}_1^{(i)}&-\mathfrak{s}_2^{(i)}\\ i\mathfrak{s}_2^{(i)}&i\mathfrak{s}_3^{(i)}&-i\mathfrak{s}_2^{*(i)}&i\mathfrak{s}_1^{(i)}\end{pmatrix},\tag{3.14}$$

where $\mathfrak{s}_1^{(i)}$ ($\mathfrak{s}_1^{(i)}\neq 0$) and $\mathfrak{s}_3^{(i)}$ are pure imaginary numbers, while $\mathfrak{s}_2^{(i)}$ is uncertain. On the other hand, by (3.13) we have

$$\rho(k)=\lim_{i\to\infty}\rho^{(i)}(k),\qquad r_1(k)=\lim_{i\to\infty}r_1^{(i)}(k),\qquad r_2(k)=\lim_{i\to\infty}r_2^{(i)}(k).\tag{3.15}$$

From (3.14), we know that for each i, $\rho^{(i)}(k)$, $r_1^{(i)}(k)$, and $r_2^{(i)}(k)$ have well-defined nonzero limits as $k\to 0$. It follows from (3.15) that $\rho(k)$, $r_1(k)$, and $r_2(k)$ are well-behaved at the origin by the dominated convergence.

Next, we prove the statement (d). Recall symmetries (2.35) and (2.49), for $k\in\mathbb{R}_+e^{\frac{\pi}{4}i}$ we have

$$s_{12}^{(i)}(k)=s_{32}^{*(i)}(k^*)=s_{21}^{*(i)}(ik^*)=s_{21}^{*(i)}(k);\qquad s_{11}^{(i)}(k)=s_{33}^{*(i)}(k^*)=s_{22}^{*(i)}(ik^*)=s_{22}^{*(i)}(k).\tag{3.16}$$

Then we conclude that

$$1-|r_1^{(i)}(k)|^2=1-\frac{s_{12}^{(i)}(k)s_{21}^{(i)}(k)}{s_{11}^{(i)}(k)s_{22}^{(i)}(k)}=\frac{s_{11}^{(i)}(k)s_{22}^{(i)}(k)-s_{12}^{(i)}(k)s_{21}^{(i)}(k)}{|s_{22}^{(i)}(k)|^2}=\frac{\Delta^{(i)}(k)}{|s_{22}^{(i)}(k)|^2}.\tag{3.17}$$

Letting $i\to\infty$ in the above equation gives

$$1-|r_1(k)|^2=\frac{\Delta(k)}{|s_{22}(k)|^2},\qquad k\in\mathbb{R}_+e^{\frac{\pi}{4}i}.\tag{3.18}$$

Since the left-hand side of (3.18) is real and tends to 1 as $|k| \to \infty$, and since the right-hand side is nonzero for all $k \in \mathbb{R}_+ e^{\frac{\pi}{4}i}$ by Assumption 3.1, we deduce that $1 - |r_1(k)|^2 > 0$ for all $k \in \mathbb{R}_+ e^{\frac{\pi}{4}i}$. An analogous argument shows that $|r_2(k)| < 1$ for $k \in \mathbb{R}_+ e^{\frac{\pi}{4}i}$.

It remains to prove $|\rho(k)| < 1$. Again by symmetries (2.35), we have

$$(\Delta^{(i)}(k))^* = \det \begin{pmatrix} s_{33}^{(i)} & s_{32}^{(i)} \\ s_{23}^{(i)} & s_{22}^{(i)} \end{pmatrix}(k), \quad (\Lambda^{(i)}(k))^* = \det \begin{pmatrix} s_{32}^{(i)} & s_{31}^{(i)} \\ s_{22}^{(i)} & s_{21}^{(i)} \end{pmatrix}(k), \quad k \in \mathbb{R}_+. \tag{3.19}$$

Then direct calculation shows that

$$1 - |\rho^{(i)}(k)|^2 = 1 - \frac{\Lambda^{(i)}(k)(\Lambda^{(i)}(k))^*}{\Delta^{(i)}(k)(\Delta^{(i)}(k))^*} = \frac{\Delta^{(i)}(k)(\Delta^{(i)}(k))^* - \Lambda^{(i)}(k)(\Lambda^{(i)}(k))^*}{|\Delta^{(i)}(k)|^2} = \frac{s_{22}^{(i)}(k) s_{44}^{A(i)}(k)}{|\Delta^{(i)}(k)|^2}. \tag{3.20}$$

Let $i \to \infty$, then we get

$$1 - |\rho(k)|^2 = \frac{s_{22}(k) s_{44}^{A}(k)}{|\Delta(k)|^2}, \qquad k \in \mathbb{R}_+. \tag{3.21}$$

Using Assumption 3.1 again, the same discussion as above gives $|\rho(k)| < 1$ for $k > 0$. □

Remark 11. In fact, it is easy to verify that r_2 and r_1 satisfy the relation $r_2(k) = -\mathrm{i} r_1(k)$ by (2.49) and (2.35). Consequently, we immediately obtain $|r_2(k)| < 1$ for $k \in \mathbb{R}_+ e^{\frac{\pi}{4}i}$.

Remark 12. Let $k \to 0$ on both sides of (3.21), we find that $|\rho(0)| = 1$.

3.1.1 | The RH Problem for $M(x, t, k)$: Soliton-Free Case

In this subsection, we define an appropriate RH problem, and conclude that the fundamental meromorphic eigenfunction $M(x, t, k)$ satisfies the RH problem under our assumptions. The RH problem is defined as follows.

RH Problem 1. Seek a 4×4 matrix-valued function $M(x, t, k)$ satisfying the following conditions:

- $M(x, t, k)$ is analytic for $k \in \mathbb{C} \setminus \Gamma$.
- When k approaches Γ from the left or right, the limits of $M(x, t, k)$ exist, denoted by $M_+(x, t, k)$ and $M_-(x, t, k)$. Furthermore, $M_+(x, t, k)$ and $M_-(x, t, k)$ are related by

$$M_+(x, t, k) = M_-(x, t, k) e^{\Theta(x,t,k)} v(k) e^{-\Theta(x,t,k)}, \qquad k \in \Gamma \setminus \{0\}. \tag{3.22}$$

- $M(x, t, k)$ has the following two symmetries:

$$M(x, t, k) = A M(x, t, \mathrm{i}k) A^{-1}, \quad M(x, t, k) = B M^*(x, t, k^*) B, \tag{3.23}$$

  where A and B are defined by (2.12).
- As $k \to \infty$

$$M(x, t, k) = I + \frac{M^{(1)}(x, t)}{k} + \frac{M^{(2)}(x, t)}{k^2} + \mathcal{O}\left(\frac{1}{k^3}\right), \tag{3.24}$$

  where the matrices $M^{(1)}(x, t)$ and $M^{(2)}(x, t)$ satisfy

$$M_{12}^{(1)}(x, t) = M_{13}^{(1)}(x, t) = 0. \tag{3.25}$$

- There exist matrices $\left\{\mathcal{M}_1^{(l)}\right\}_{l=-1}^{p}$ such that for any $N \geq -1$

$$M_1(x,k) = \sum_{l=-1}^{N} \mathcal{M}_1^{(l)}(x,t)k^l + \mathcal{O}(k^{N+1}), \quad as \quad k \to 0, \quad k \in D_1. \tag{3.26}$$

  Moreover, $\mathcal{M}_1^{(-1)}(x,t)$ has the form:

$$\mathcal{M}_1^{(-1)}(x,t) = \begin{pmatrix} -\mathrm{i}\delta_1 & -\mathrm{i}\alpha_+ & 0 & 0 \\ \delta_2 & -\beta_+ & 0 & 0 \\ \mathrm{i}\delta_1 & \mathrm{i}\alpha_+ & 0 & 0 \\ -\delta_2 & \beta_+ & 0 & 0 \end{pmatrix},$$

  where $\alpha_+(x,t)$, $\beta_+(x,t)$, $\delta_1(x,t)$, and $\delta_2(x,t)$ are scalar functions of x and t.
- $M(x,t,k)$ and $M^A(x,t,k)$ satisfy

$$JM(x,t,-k)J^{-1} = M^A(x,t,k), \quad k \in \mathbb{C} \setminus \Gamma, \tag{3.27}$$

  where $M^A(x,t,k) = (M^{-1}(x,t,k))^\top$.

In the following, we will prove the uniqueness of the solution of RH problem 1. We first verify that any solution of the RH problem 1 has a unit determinant.

Lemma 3.6. *Suppose that $M(x,t,k)$ is the solution of RH problem 1. Then* $\det M(x,t,k) = 1$.

Proof. Note that $\det M$ has no jump across $\Gamma \setminus \{0\}$, so $\det M$ is analytic for $k \in \mathbb{C} \setminus \{0\}$. On the one hand, it is easy to deduce from the asymptotic behavior (3.26) that $\det M$ has at most a double pole at 0. That is, there exist functions $\tilde{g}_0$, $\tilde{g}_1$, and $\tilde{g}_2$ such that

$$\det M = \tilde{g}_0 + \frac{\tilde{g}_1}{k} + \frac{\tilde{g}_2}{k^2}. \tag{3.28}$$

On the other hand, from the symmetry (3.23) satisfied by $M(x,t,k)$, it is obtained that $M^{(1)}(x,t)$ and $M^{(2)}(x,t)$ in Equation (3.24) must have the following form:

$$M^{(1)}(x,t) = \begin{pmatrix} M_{11}^{(1)} & M_{12}^{(1)} & M_{13}^{(1)} & -M_{12}^{*(1)} \\ \mathrm{i}M_{12}^{*(1)} & -\mathrm{i}M_{11}^{(1)} & -\mathrm{i}M_{12}^{(1)} & -\mathrm{i}M_{13}^{(1)} \\ -M_{13}^{(1)} & M_{12}^{*(1)} & -M_{11}^{(1)} & -M_{12}^{(1)} \\ \mathrm{i}M_{12}^{(1)} & \mathrm{i}M_{13}^{(1)} & -\mathrm{i}M_{12}^{*(1)} & \mathrm{i}M_{11}^{(1)} \end{pmatrix}, \tag{3.29}$$

$$M^{(2)}(x,t) = \begin{pmatrix} M_{11}^{(2)} & M_{12}^{(2)} & M_{13}^{(2)} & M_{12}^{*(2)} \\ -M_{12}^{*(2)} & -M_{11}^{(2)} & -M_{12}^{(2)} & -M_{13}^{(2)} \\ M_{13}^{(2)} & M_{12}^{*(2)} & M_{11}^{(2)} & M_{12}^{(2)} \\ -M_{12}^{(2)} & -M_{13}^{(2)} & -M_{12}^{*(2)} & -M_{11}^{(2)} \end{pmatrix}, \tag{3.30}$$

where $M_{11}^{*(1)} = -M_{11}^{(1)}$, $M_{13}^{*(1)} = -M_{13}^{(1)}$, $M_{11}^{(2)} = M_{11}^{*(2)}$, and $M_{13}^{*(2)} = -M_{13}^{(2)}$. By condition (3.25), as $k \to \infty$ we have

$$M(x,t,k) = I + \frac{1}{k}\begin{pmatrix} M_{11}^{(1)} & 0 & 0 & 0 \\ 0 & -\mathrm{i}M_{11}^{(1)} & 0 & 0 \\ 0 & 0 & -M_{11}^{(1)} & 0 \\ 0 & 0 & 0 & \mathrm{i}M_{11}^{(1)} \end{pmatrix} + \frac{1}{k^2}\begin{pmatrix} M_{11}^{(2)} & M_{12}^{(2)} & M_{13}^{(2)} & M_{12}^{*(2)} \\ -M_{12}^{*(2)} & -M_{11}^{(2)} & -M_{12}^{(2)} & -M_{13}^{(2)} \\ M_{13}^{(2)} & M_{12}^{*(2)} & M_{11}^{(2)} & M_{12}^{(2)} \\ -M_{12}^{(2)} & -M_{13}^{(2)} & -M_{12}^{*(2)} & -M_{11}^{(2)} \end{pmatrix} + \mathcal{O}\left(\frac{1}{k^3}\right). \tag{3.31}$$

From this asymptotic expansion, we find that $\det M = 1 + \mathcal{O}\left(\frac{1}{k^3}\right)$. Combining this with (3.28), we immediately get $\tilde{g}_0 = 1$ and $\tilde{g}_1 = \tilde{g}_2 = 0$. □

Lemma 3.7. *If RH problem 1 has a solution, then the solution is unique.*

Proof. Suppose M and N are two solutions of RH problem 1. By Lemma 3.6, we know $\det M = \det N = 1$, so M and N are invertible. By Equation (3.27), we obtain $N^A(x,t,k) = JN(x,t,-k)J^{-1} = JA^2N(x,t,k)A^2J^{-1}$. Note that $N^{-1}(x,t,k) = (N^A(x,t,k))^\top$, then it is not difficult to obtain

$$N^{-1}(x,t,k) = \frac{1}{k}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -\mathrm{i}\delta_1 & \mathrm{i}\delta_2 & -\mathrm{i}\delta_1 & \mathrm{i}\delta_2 \\ -\alpha_+ & -\beta_+ & -\alpha_+ & -\beta_+ \end{pmatrix} + \mathcal{O}(1), \qquad k \to 0, \quad k \in D_1 \tag{3.32}$$

by using the asymptotic expansion (3.26). Therefore, direct calculation shows that

$$MN^{-1} = \mathcal{O}\left(\frac{1}{k}\right), \qquad k \to 0, \quad k \in D_1. \tag{3.33}$$

Moreover, it is easy to see that MN^{-1} is analytic for $k \in \mathbb{C} \setminus \{0\}$. Then by asymptotic behavior (3.33), we know that there exists two matrix-valued functions G_1 and G_2 such that

$$MN^{-1} = G_1 + \frac{G_2}{k}. \tag{3.34}$$

On the other hand, by (3.31) and the equation

$$N^A(x,t,k) = JA^2N(x,t,k)A^2J^{-1},$$

it is easy to see that

$$M(x,t,k) = I + \frac{M_{11}^{(1)}(x,t)}{k}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & \mathrm{i} \end{pmatrix} + \mathcal{O}\left(\frac{1}{k^2}\right), \qquad k \to \infty,$$

$$N^{-1}(x,t,k) = I - \frac{N_{11}^{(1)}(x,t)}{k}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & \mathrm{i} \end{pmatrix} + \mathcal{O}\left(\frac{1}{k^2}\right), \qquad k \to \infty. \tag{3.35}$$

So we have

$$MN^{-1} = I + \frac{M_{11}^{(1)} - N_{11}^{(1)}}{k}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & \mathrm{i} \end{pmatrix} + \mathcal{O}\left(\frac{1}{k^2}\right), \qquad k \to \infty. \tag{3.36}$$

Combining (3.36) with (3.34), we immediately get

$$MN^{-1} = I + \frac{M_{11}^{(1)} - N_{11}^{(1)}}{k}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & \mathrm{i} \end{pmatrix}. \tag{3.37}$$

Note that $\det(MN^{-1}) = 1$, so there must be $M_{11}^{(1)} - N_{11}^{(1)} = 0$, that is, $M = N$. □

Combining the results of Section 2.6 and this section, we conclude the following theorem.

Theorem 3.2. *Suppose $\{u(x,t), v(x,t)\}$ is a Schwartz class solution of (1.1) with existence time $T \in (0,\infty]$ and initial data u_0, v_0 such that Assumptions 3.1 and 3.2 hold. Then the RH problem 1 has a unique solution $M(x,t,k)$ for each $\mathbb{R}\times[0,T)$. Moreover, the corresponding solution $\{u,v\}$ is reconstructed as*

$$u(x,t) = 2\partial_x \lim_{k\to\infty} k(M_{44}(x,t,k)-1) = 4\lim_{k\to\infty} k^2(M_{13}(x,t,k)), \tag{3.38}$$

$$v(x,t) = 4\mathrm{Im}(\lim_{k\to\infty} k^2(M_{12}(x,t,k))), \tag{3.39}$$

$$3v^2(x,t) = 2\partial_t \lim_{k\to\infty} k(M_{44}(x,t,k)-1) - \frac{1}{2}u_{xx} + \frac{3}{2}u^2. \tag{3.40}$$

Proof. Under the assumption of the theorem, the fundamental meromorphic eigenfunction $M(x,t,k)$ exists, and it is easy to verify that it satisfies RH problem 1. Then by the uniqueness result (see Lemma 3.7), we know that $M(x,t,k)$ is the unique solution of RH problem 1. The reconstruction formulas (3.38) and (3.39) can be obtained by calculating the asymptotic behavior of M as $k\to\infty$ (see (3.52) below). Finally, taking the reconstruction formula (3.38) of u into (1.1), we get (3.40). □

3.2 | Pure Soliton Case

In this subsection, we consider the case that the reflection coefficients are always zero, that is, only the discrete data contribute to the potentials. It is well-known that this setting will enable us to obtain explicit soliton solutions. The set of discrete spectrum is defined as all poles of $M(x,t,k)$ except the origin, that is, the set $\mathcal{Z}$ defined in Section 2.6. We need add some appropriate restrictions to the discrete spectrum. Under these conditions, we define an RH problem, and we show that the soliton solutions of (1.1) can be constructed from the solution of the RH problem.

3.2.1 | Pure Soliton RH Problem

We first note that in this case, the asymptotic behavior of $M(x,t,k)$ as $k\to 0$ differs significantly from that in the soliton-free case, since Assumptions 3.1 and 3.2 no longer hold. Therefore, we need to reexamine the asymptotic behavior of $M(x,t,k)$ as $k\to 0$. Since M has no jump on Γ in this case, $M(x,t,k)$ admits the following Laurent expansion near the origin:

$$M(x,t,k) = \sum_{l=-1}^{\infty} \mathcal{M}^{(l)}(x,t)k^l. \tag{3.41}$$

Due to symmetry (3.46), $\mathcal{M}^{(-1)}$ in Equation (3.41) must satisfy

$$A(-\mathrm{i}\mathcal{M}^{(-1)})A^{-1} = \mathcal{M}^{(-1)}, \quad B\mathcal{M}^{*(-1)}B = \mathcal{M}^{(-1)}. \tag{3.42}$$

In addition, we note that $[M_1]_2 = \mu_{+2}$. Then combining (3.42) with the asymptotic expansion of μ_{+2} as $k\to 0$ (see (2.23)), it is not difficult to find that $\mathcal{M}^{(-1)}$ has the form

$$\mathcal{M}^{(-1)} = \tilde{\delta}_1\begin{pmatrix} \mathrm{i} & 0 & \mathrm{i} & 0\\ 0 & 1 & 0 & 1\\ -\mathrm{i} & 0 & -\mathrm{i} & 0\\ 0 & -1 & 0 & -1\end{pmatrix} + \tilde{\delta}_2\begin{pmatrix} 0 & \mathrm{i} & 0 & \mathrm{i}\\ 1 & 0 & 1 & 0\\ 0 & -\mathrm{i} & 0 & -\mathrm{i}\\ -1 & 0 & -1 & 0\end{pmatrix}, \tag{3.43}$$

where $\tilde{\delta}_1$ and $\tilde{\delta}_2$ are some scalar functions of x and t.

In order to complete the formulation of the RH problem, one has to determine the residue conditions at the possible poles of $M(x,t,k)$. We have failed to directly derive the corresponding residue conditions, even if we have clarified the specific form of $M(x,t,k)$ (see (3.12) for M_1). However, we find that if the following residue conditions are imposed on $M(x,t,k)$, we will finally get the soliton solutions. We next describe our setting in detail.

First, we assume that $M(x,t,k)$ has only simple poles. Moreover, we restrict all the poles of $M_1(x,t,k)$ to the ray $\mathbb{R}_+ e^{\frac{\pi i}{4}}$ and denote them as $\mathcal{Z}_1 = \{\zeta_j\}_{j=1}^N$. Then all the discrete spectrum can be determined by $\mathcal{Z}_1$ according to A- and B-symmetry. Next, we define the residue function $F_j(x,t)$ associated with ζ_j by

$$F_j(x,t) = \tau_j e^{(\theta_2-\theta_3)(x,t,\zeta_j)}, \tag{3.44}$$

where $\tau_j > 0$ and $\theta_i(x,t,k) = \Theta_{ii}(x,t,k)$ for $i = 1,2,3,4$. We then require $M(x,t,k)$ to satisfy the following residue conditions:

$$\begin{aligned}
Res_{k=\zeta_j} M(x,t,k) &= \lim_{k\to\zeta_j} M(x,t,k)\begin{pmatrix} 0 & 0 & 0 & iF_j \\ 0 & 0 & F_j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \\
Res_{k=i\zeta_j} M(x,t,k) &= \lim_{k\to i\zeta_j} M(x,t,k)\begin{pmatrix} 0 & iF_j & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -F_j & 0 \end{pmatrix}, \\
Res_{k=-\zeta_j} M(x,t,k) &= \lim_{k\to-\zeta_j} M(x,t,k)\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -iF_j & 0 & 0 \\ -F_j & 0 & 0 & 0 \end{pmatrix}, \\
Res_{k=-i\zeta_j} M(x,t,k) &= \lim_{k\to-i\zeta_j} M(x,t,k)\begin{pmatrix} 0 & 0 & 0 & 0 \\ F_j & 0 & 0 & 0 \\ 0 & 0 & 0 & -iF_j \\ 0 & 0 & 0 & 0 \end{pmatrix}.
\end{aligned} \tag{3.45}$$

If the fundamental meromorphic eigenfunction $M(x,t,k)$ satisfies our above assumptions on discrete data, then it will satisfy the following RH problem:

RH Problem 2 (Pure soliton RH problem). Seek a 4×4 matrix-valued function $M(x,t,k)$ satisfying the following conditions:

- $M(x,t,k)$ is analytic for $k \in \mathbb{C}\setminus\{0\cup\mathcal{Z}\}$, where $\mathcal{Z} = \pm\mathcal{Z}_1 \cup \pm i\mathcal{Z}_1$.
- $M(x,t,k)$ has the following two symmetries:

$$M(x,t,k) = AM(x,t,ik)A^{-1}, \quad M(x,t,k) = BM^*(x,t,k^*)B, \tag{3.46}$$

  where A and B are defined by (2.12).
- As $k\to\infty$

$$M(x,t,k) = I + \frac{M^{(1)}(x,t)}{k} + \frac{M^{(2)}(x,t)}{k^2} + \mathcal{O}\left(\frac{1}{k^3}\right), \tag{3.47}$$

  where the matrices $M^{(1)}(x,t)$ and $M^{(2)}(x,t)$ satisfy

$$M_{12}^{(1)}(x,t) = M_{13}^{(1)}(x,t) = 0, \qquad M_{13}^{(2)}(x,t) = \frac{1}{2}\partial_x(M_{44}^{(1)}(x,t)). \tag{3.48}$$

- $M(x,t,k)$ admits the following Laurent expansion near the origin:

$$M(x,t,k) = \sum_{l=-1}^{\infty} \mathcal{M}^{(l)}(x,t)k^l, \tag{3.49}$$

where $\mathcal{M}^{(-1)}$ has the form

$$\mathcal{M}^{(-1)} = \tilde{\delta}_1 \begin{pmatrix} \mathrm{i} & 0 & \mathrm{i} & 0 \\ 0 & 1 & 0 & 1 \\ -\mathrm{i} & 0 & -\mathrm{i} & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix} + \tilde{\delta}_2 \begin{pmatrix} 0 & \mathrm{i} & 0 & \mathrm{i} \\ 1 & 0 & 1 & 0 \\ 0 & -\mathrm{i} & 0 & -\mathrm{i} \\ -1 & 0 & -1 & 0 \end{pmatrix}, \tag{3.50}$$

where $\tilde{\delta}_1$ and $\tilde{\delta}_2$ are some scalar functions of x and t.

- $M(x,t,k)$ and $M^A(x,t,k)$ satisfy

$$JM(x,t,-k)J^{-1} = M^A(x,t,k), \quad k \in \mathbb{C} \setminus \{0 \cup \mathcal{Z}\}, \tag{3.51}$$

where $M^A(x,t,k) = (M^{-1}(x,t,k))^\top$.

- $M(x,t,k)$ has simple poles at each point of $\mathcal{Z}$, and the residue conditions are given by (3.45).

Remark 13. The fundamental meromorphic eigenfunctions $M(x,t,k)$ has the following asymptotic behavior:

$$M(x,t,k) = I + \frac{\mu_+^{[1]}(x,t)}{k} + \frac{\mu_+^{[2]}(x,t)}{k^2} + \mathcal{O}\left(\frac{1}{k^3}\right), \qquad k \to \infty, \tag{3.52}$$

where $\mu_+^{[1]}(x,t)$ and $\mu_+^{[2]}(x,t)$ have the same form as $\mu_+^{[1]}(x)$ and $\mu_+^{[2]}(x)$. Therefore, the eigenfunctions $M(x,t,k)$ satisfy the second equation in (3.48). This equation will be used in the proof of Theorem 3.3.

Next we will explain the rationality of the residue conditions we set. Now let us forget how to construct $M(x,t,k)$ and how to use $M(x,t,k)$ to derive the RH problem. We hope to show that, in turn, from this RH problem, we can reconstruct the solutions of system (1.1). In this sense, it is appropriate for us to impose these residue conditions. But before that, we have to give the uniqueness result of the solution of RH problem 2. We can prove this by taking the same steps as proving the uniqueness of the solution of RH problem 1. We briefly summarize this proof. Suppose that M and $\tilde{M}$ are two different solutions of the RH problem. We first prove that any solution of RH problem 2 has a unit determinant as shown in Lemma 3.6. Then, we only need to prove that $M^{-1}\tilde{M} = I$. Therefore, we have to consider the asymptotic behavior of $M^{-1}\tilde{M}$ as $k \to 0$. From symmetry (3.51) and asymptotic behavior (3.53), we can deduce that M^{-1} has the asymptotic behavior:

$$M^{-1}(x,t,k) = \sum_{l=-1}^{\infty} \mathcal{N}^{(l)}(x,t)k^l, \qquad k \to 0, \tag{3.53}$$

where $\mathcal{N}^{(-1)}$ has the form:

$$\mathcal{N}^{(-1)} = -\tilde{\delta}_1 \begin{pmatrix} \mathrm{i} & 0 & \mathrm{i} & 0 \\ 0 & 1 & 0 & 1 \\ -\mathrm{i} & 0 & -\mathrm{i} & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix} - \tilde{\delta}_2 \begin{pmatrix} 0 & \mathrm{i} & 0 & \mathrm{i} \\ 1 & 0 & 1 & 0 \\ 0 & -\mathrm{i} & 0 & -\mathrm{i} \\ -1 & 0 & -1 & 0 \end{pmatrix}. \tag{3.54}$$

Thus $M^{-1}\tilde{M} = O\left(\frac{1}{k}\right)$ as $k \to 0$. Now the key to check whether the proof of Lemma 3.7 is still valid is to verify that $\{\zeta_j\}_{j=1}^N$ are removable singularities of $M\tilde{M}^{-1}$. One can prove this fact by the method we use in Theorem 3.3.
Now we are ready to show how to reconstruct potentials from the solution of RH problem 2.

Theorem 3.3. *Suppose that $M(x,t,k)$ solves the Pure soliton RH problem. Then the functions $\{u(x,t), v(x,t)\}$ defined by*

$$u(x,t) = 2\partial_x \lim_{k\to\infty} k(M_{44}(x,t,k) - 1), \qquad v(x,t) = 4\mathrm{Im}(\lim_{k\to\infty} k^2(M_{12}(x,t,k))) \tag{3.55}$$

satisfy HScKdV system (1.1). Moreover $v(x,t)$ can also be expressed as

$$3v^2(x,t) = 2\partial_t \lim_{k\to\infty} k(M_{44}(x,t,k) - 1) - \frac{1}{2}u_{xx} + \frac{3}{2}u^2. \tag{3.56}$$

Proof. In the spirit of the well-known result of Zakharov and Shabat [33], we can verify these reconstruction formulas. We first define a matrix-valued function $\psi(x,t,k)$ by

$$\psi(x,t,k) = P(k)M(x,t,k)\mathrm{e}^{\Theta(x,t,k)}. \tag{3.57}$$

It is easy to see that $\psi_x\psi^{-1}$ is analytic for $k \in \mathbb{C} \setminus \{0 \cup \mathcal{Z}\}$. Moreover, we have

$$\begin{aligned}\psi_x\psi^{-1} &= P(M_x + ML)\mathrm{e}^{\Theta}\mathrm{e}^{-\Theta}M^{-1}P^{-1}\\ &= PM_xM^{-1}P^{-1} + PMLM^{-1}P^{-1}.\end{aligned} \tag{3.58}$$

According to the asymptotic behavior (3.50), (3.53), the expressions of P and P^{-1}, and the symmetry (3.46), one can verify that $\psi_x\psi^{-1}$ is $\mathcal{O}(1)$ as $k \to 0$. Therefore, $\psi_x\psi^{-1}$ is analytic except for $k \in \mathcal{Z}$. We next show that the points in set $\mathcal{Z}$ are removable singularities of $\psi_x\psi^{-1}$, and then $\psi_x\psi^{-1}$ is holomorphic on the entire complex plane. To this end, we define

$$W(x,t,k) = P^{-1}(k)\psi(x,t,k) = M(x,t,k)\mathrm{e}^{\Theta(x,t,k)}. \tag{3.59}$$

It is enough to verify that these points are removable singularities of W_xW^{-1}. Noting that $W(x,t,k)$ satisfies symmetries $AW(x,t,\mathrm{i}k)A^{-1} = W(x,t,k)$ and $BW^*(x,t,k^*)B = W(x,t,k)$, we only need to verify that $\{\zeta_j\}_{j=1}^N$ are removable singularities of W_xW^{-1}. A direct calculation shows that $W(x,t,k)$ satisfies the following residue conditions at ζ_j and $-\zeta_j$:

$$\mathrm{Res}_{k=\zeta_j}W(x,t,k) = \lim_{k\to\zeta_j} W(x,t,k)\begin{pmatrix}0&0&0&\mathrm{i}\tau_j\\0&0&\tau_j&0\\0&0&0&0\\0&0&0&0\end{pmatrix}, \tag{3.60}$$

$$\mathrm{Res}_{k=-\zeta_j}W(x,t,k) = \lim_{k\to-\zeta_j} W(x,t,k)\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&-\mathrm{i}\tau_j&0&0\\-\tau_j&0&0&0\end{pmatrix}. \tag{3.61}$$

By differentiation, it is easy to check that $W_x(x,t,k)$ satisfies the same residue condition with $W(x,t,k)$. Therefore, we assume that the Laurent expansions of W and W_x at ζ_j and $-\zeta_j$, respectively, have the following forms:

$$W_x(x,t,k) = \frac{1}{k-\zeta_j}\begin{pmatrix}0&0&\tau_jB_1&\mathrm{i}\tau_jA_1\\0&0&\tau_jB_2&\mathrm{i}\tau_jA_2\\0&0&\tau_jB_3&\mathrm{i}\tau_jA_3\\0&0&\tau_jB_4&\mathrm{i}\tau_jA_4\end{pmatrix} + \begin{pmatrix}A_1&B_1&C_1&D_1\\A_2&B_2&C_2&D_2\\A_3&B_3&C_3&D_3\\A_4&B_4&C_4&D_4\end{pmatrix} + \mathcal{O}(k-\zeta_j), \tag{3.62}$$

$$W(x,t,k) = \frac{1}{k+\zeta_j}\begin{pmatrix}-\tau_jd_1&-\mathrm{i}\tau_jc_1&0&0\\-\tau_jd_2&-\mathrm{i}\tau_jc_2&0&0\\-\tau_jd_3&-\mathrm{i}\tau_jc_3&0&0\\-\tau_jd_4&-\mathrm{i}\tau_jc_4&0&0\end{pmatrix} + \begin{pmatrix}a_1&b_1&c_1&d_1\\a_2&b_2&c_2&d_2\\a_3&b_3&c_3&d_3\\a_4&b_4&c_4&d_4\end{pmatrix} + \mathcal{O}(k+\zeta_j). \tag{3.63}$$

Then by the symmetry (3.51), the definition of W (3.59) and the expansion of W (3.63), we get the Laurent expansion of W^{-1} at $k = \zeta_j$:

$$W^{-1}(x,t,k) = \frac{1}{k-\zeta_j}\begin{pmatrix}\tau_jd_1&\mathrm{i}\tau_jd_2&-\tau_jd_3&-\mathrm{i}\tau_jd_4\\\tau_jc_1&\mathrm{i}\tau_jc_2&-\tau_jc_3&-\mathrm{i}\tau_jc_4\\0&0&0&0\\0&0&0&0\end{pmatrix} + \begin{pmatrix}a_1&\mathrm{i}a_2&-a_3&-\mathrm{i}a_4\\-\mathrm{i}b_1&b_2&\mathrm{i}b_3&-b_4\\-c_1&-\mathrm{i}c_2&c_3&\mathrm{i}c_4\\\mathrm{i}d_1&-d_2&-\mathrm{i}d_3&d_4\end{pmatrix} + \mathcal{O}(k-\zeta_j). \tag{3.64}$$

Now combining (3.62) with (3.64), long but straightforward calculations show that $W_x W^{-1}$ has no negative powers of $k-\zeta_j$ in its Laurent expansion and therefore is analytic at ζ_j. Hence $\psi_x\psi^{-1}$ is a polynomial. We can determine it by analyzing its asymptotic behavior at infinity.

Next let us determine the exact form of $\psi_x\psi^{-1}$. It is easy to verify that as $k\to\infty$, M and M^{-1} have the asymptotic expansion:

$$M = I + \frac{M^{(1)}}{k} + \frac{M^{(2)}}{k^2} + \mathcal{O}\left(\frac{1}{k^3}\right), \tag{3.65}$$

$$M^{-1} = I - \frac{M^{(1)}}{k} + \frac{J^{-1}(M^{(2)})^{\top}J}{k^2} + \mathcal{O}\left(\frac{1}{k^3}\right), \tag{3.66}$$

where the exact form of $M^{(1)}$ and $M^{(2)}$ can be found in Equation (3.31). We use $[A]_d$ to represent the diagonal part of matrix A, and $[A]_0$ to represent the nondiagonal part of matrix A. Then substituting the expressions of M and M^{-1} into $PMLM^{-1}P^{-1}$ yields

$$PMLM^{-1}P^{-1} = \Pi_1 + \Pi_2 + \Pi_3 + \Pi_4 + \mathcal{O}\left(\frac{1}{k}\right), \qquad k\to\infty, \tag{3.67}$$

where

$$\Pi_1 = PLP^{-1},$$

$$\Pi_2 = P\frac{M^{(1)}}{k}LP^{-1} - P\frac{M^{(1)}}{k}LP^{-1} = 0,$$

$$\Pi_3 = \frac{1}{k}P([(M^{(2)} + (M^{(2)})^{\top}]_d - M^{(1)}M^{(1)})JP^{-1},$$

$$\Pi_4 = \frac{1}{k}P[(M^{(2)} + (M^{(2)})^{\top}]_o JP^{-1}.$$

It is easy to see that $\Pi_1 = U^{\infty}$, $\Pi_2 = 0$. Take the expansions (3.65) and (3.66) into $MM^{-1} = I$, and compare the coefficients of $\frac{1}{k^2}$ to obtain that $\Pi_3 = 0$. It now remains to calculate $PM_xM^{-1}P^{-1}$. A direct calculation shows

$$PM_xM^{-1}P^{-1} = \partial_x M_{44}^{(1)}\begin{pmatrix}0&0&0&0\\0&0&0&0\\1&0&0&0\\0&1&0&0\end{pmatrix} + \mathcal{O}\left(\frac{1}{k}\right), \qquad k\to\infty. \tag{3.68}$$

Since $\psi_x\psi^{-1}$ is an entire function on the complex plane, combining (3.67) with (3.68) we have

$$\psi_x\psi^{-1} = U^{\infty} + \partial_x M_{44}^{(1)}\begin{pmatrix}0&0&0&0\\0&0&0&0\\1&0&0&0\\0&1&0&0\end{pmatrix} \tag{3.69}$$

$$+\begin{pmatrix}0&0&0&0\\0&0&0&0\\0&-1&0&1\\\mathrm{i}&0&-\mathrm{i}&0\end{pmatrix}[M^{(2)} + (M^{(2)})^{\top}]_o J\begin{pmatrix}0&\frac{1}{2}&0&0\\\frac{1}{2}&0&0&0\\0&\frac{1}{2}&0&0\\\frac{1}{2}&0&0&0\end{pmatrix}, \tag{3.70}$$

which is reduced to

$$\psi_x\psi^{-1} = U^{\infty} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \partial_x M_{44}^{(1)} + 2M_{13}^{(2)} & 2\mathrm{i}\left(M_{12}^{*(2)} - M_{12}^{(2)}\right) & 0 & 0 \\ 2\mathrm{i}\left(M_{12}^{*(2)} - M_{12}^{(2)}\right) & \partial_x M_{44}^{(1)} + 2M_{13}^{(2)} & 0 & 0 \end{pmatrix}. \tag{3.71}$$

By (3.48), we have $M_{13}^{(2)}(x,t) = \frac{1}{2}\partial_x(M_{44}^{(1)}(x,t))$. Then we conclude that $\psi(x,t,k)$ satisfies the x-part in (2.1), where u and v are defined by (3.55). Analogous arguments give the t-part in (2.1). Finally, substituting the reconstruction formula (3.55) of u into (1.1) yields (3.56). □

Remark 14. In the above proof, we have to use the second equation in condition (3.48). From the next subsection, we will see that this condition is not important in calculating soliton solutions. But we have no way to remove the condition.

3.2.2 | Solition Solutions

Theorem 3.3 implies that if we can solve RH problem 2, then we can reconstruct the solution of Equation (1.1). However, we find that solving RH problem 2 directly is more complicated because it is a system related to a 4×4 matrix and the solution of the system has a singularity at $k = 0$. To avoid these difficulties, we introduce the row vector function $n(x,t,k)$ as follows:

$$n(x,t,k) = \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix} M(x,t,k). \tag{3.72}$$

The following lemma summarizes the basic properties of $n(x,t,k)$.

Lemma 3.8. *Assume that $M(x,t,k)$ is the solution of the Pure soliton RH problem. Then the row vector function $n(x,t,k)$ defined by (3.72) satisfies*

- *$n(x,t,k)$ is analytic for $k \in \mathbb{C} \setminus \{0 \cup \mathcal{Z}\}$.*
- *$n(x,t,k)$ has the following two symmetries:*

$$n(x,t,k) = n(x,t,\mathrm{i}k)A^{-1}, \quad n(x,t,k) = n^*(x,t,k^*)B. \tag{3.73}$$

- *As $k \to \infty$*

$$n(x,t,k) = (1,1,1,1) + \mathcal{O}\left(\frac{1}{k}\right). \tag{3.74}$$

- *$n(x,t,k)$ is $\mathcal{O}(1)$ as $k \to 0$.*
- *$n(x,t,k)$ has simple poles at the points ζ_j, $\mathrm{i}\zeta_j$, $-\zeta_j$, and $-\mathrm{i}\zeta_j$, and the residue conditions are given by*

$$Res_{k=\zeta_j} n(x,t,k) = \lim_{k\to\zeta_j} n(x,t,k) \begin{pmatrix} 0 & 0 & 0 & \mathrm{i}F_j \\ 0 & 0 & F_j & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$Res_{k=\mathrm{i}\zeta_j} n(x,t,k) = \lim_{k\to\mathrm{i}\zeta_j} n(x,t,k) \begin{pmatrix} 0 & \mathrm{i}F_j & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -F_j & 0 \end{pmatrix},$$

$$Res_{k=-\zeta_j} n(x,t,k) = \lim_{k\to-\zeta_j} n(x,t,k)\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -\mathrm{i}F_j & 0 & 0 \\ -F_j & 0 & 0 & 0 \end{pmatrix},$$

$$Res_{k=-\mathrm{i}\zeta_j} n(x,t,k) = \lim_{k\to-\mathrm{i}\zeta_j} n(x,t,k)\begin{pmatrix} 0 & 0 & 0 & 0 \\ F_j & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mathrm{i}F_j \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{3.75}$$

- *Let* $n(x,t,k) = \big(n_1(x,t,k) \quad n_2(x,t,k) \quad n_3(x,t,k) \quad n_4(x,t,k)\big)$. *Then the solutions* $\{u,v\}$ *of system (1.1) can be reconstructed as follows:*

$$u(x,t) = 2\partial_x \lim_{k\to\infty} k(n_4(x,t,k)-1), \tag{3.76}$$

$$3v^2(x,t) = 2\partial_t \lim_{k\to\infty} k(n_4(x,t,k)-1) - \frac{1}{2}u_{xx} + \frac{3}{2}u^2. \tag{3.77}$$

Proof. The proof of this lemma is almost obvious. We emphasize that the new reconstruction formulas (3.76) and (3.77) hold because $M^{(1)}$ in (3.47) is a diagonal matrix. □

Now we are ready to use $n(x,t,k)$ to calculate the soliton solutions. One can see that the statements in Lemma 3.8 essentially constitute a linear system that can be solved exactly. We show below how to calculate $n(x,t,k)$. We take $N=1$ as an example. Suppose there is only one discrete spectrum $\zeta_1 := \kappa_1 e^{\frac{\pi\mathrm{i}}{4}}$ with $\kappa_1 > 0$. When setting the residue conditions, we require the constant $\tau_1 > 0$. We note that the case $\tau_1 < 0$ corresponds to singular soliton solutions of the HScKdV system. We restrict our analysis to the case $\tau_1 > 0$ here. The residue conditions (3.75) together with the normalization condition (3.74) imply that

$$\begin{aligned} n_4(x,t,k) &= 1 + \frac{-\mathrm{i}F_1}{k+\mathrm{i}\zeta_1} n_3(x,t,-\mathrm{i}\zeta_1) + \frac{\mathrm{i}F_1}{k-\zeta_1} n_1(x,t,\zeta_1) \\ &= 1 + \frac{-\mathrm{i}F_1}{k+\mathrm{i}\zeta_1} n_3(x,t,-\mathrm{i}\zeta_1) + \frac{\mathrm{i}F_1}{k-\mathrm{i}\zeta_1^*} n_1(x,t,\mathrm{i}\zeta_1^*). \end{aligned} \tag{3.78}$$

According to symmetries (3.73), we have

$$n_3(x,t,-\mathrm{i}\zeta_1) = n_4(x,t,-\zeta_1), \qquad n_1(x,t,\mathrm{i}\zeta_1^*) = n_4(x,t,-\zeta_1^*).$$

Then $n_4(x,t,k)$ can be written as

$$n_4(x,t,k) = 1 + \frac{-\mathrm{i}F_1}{k+\mathrm{i}\zeta_1} n_4(x,t,-\zeta_1) + \frac{\mathrm{i}F_1}{k-\mathrm{i}\zeta_1^*} n_4(x,t,-\zeta_1^*). \tag{3.79}$$

In Equation (3.79), let k take $-\zeta_1$ and $-\zeta_1^*$, then we get the linear algebraic system of $X := (n_4(x,t,-\zeta_1), \quad n_4(x,t,-\zeta_1^*))^\top$, namely,

$$X = \begin{pmatrix} \dfrac{-\mathrm{i}F_1}{-\zeta_1+\mathrm{i}\zeta_1} & \dfrac{\mathrm{i}F_1}{-\zeta_1-\mathrm{i}\zeta_1^*} \\ \dfrac{-\mathrm{i}F_1}{-\zeta_1^*+\mathrm{i}\zeta_1} & \dfrac{\mathrm{i}F_1}{-\zeta_1^*-\mathrm{i}\zeta_1^*} \end{pmatrix} X + \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{3.80}$$

Thus $n_4(x,t,k)$ can be determined, and by symmetries (3.73) one can determine $n(x,t,k)$ completely. Next we show how to get the soliton solutions from $n(x,t,k)$. By reconstruction formula (3.76) and Equation (3.79), $u(x,t)$ can be expressed as

$$u(x,t) = 2\partial_x\big(\mathrm{i}F_1 \cdot n_4(x,t,-\zeta_1^*) - \mathrm{i}F_1 \cdot n_4(x,t,-\zeta_1)\big). \tag{3.81}$$

Solving the algebraic system (3.80) and substituting its solution into (3.81), we obtain

$$u(x,t) = -4\kappa_1^2 \operatorname{sech}^2\left(\sqrt{2}\kappa_1\left(x - 2\kappa_1^2 t\right) - x_0\right), \tag{3.82}$$

where $x_0 = \ln(\frac{\sqrt{\tau_1}}{2\kappa_1}) \in \mathbb{R}$. Substituting the expression (3.82) of $u(x,t)$ into the formula (3.77), we have

$$v(x,t) = \pm 2\sqrt{2}\kappa_1^2 \operatorname{sech}\left(\sqrt{2}\left(x - 2\kappa_1^2 t\right) - x_0\right). \tag{3.83}$$

Then we obtain one soliton solution of the HScKdV system (1.1)

$$\begin{cases} u(x,t) = -4\kappa_1^2 \operatorname{sech}^2\left(\sqrt{2}\kappa_1\left(x - 2\kappa_1^2 t\right) - x_0\right), \\ v(x,t) = \pm 2\sqrt{2}\kappa_1^2 \operatorname{sech}\left(\sqrt{2}\kappa_1\left(x - 2\kappa_1^2 t\right) - x_0\right), \end{cases}$$

which is consistent with the form given in [1].

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Data Availability Statement

Data sharing not applicable to this article as no data sets were generated or analyzed during the current study.

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